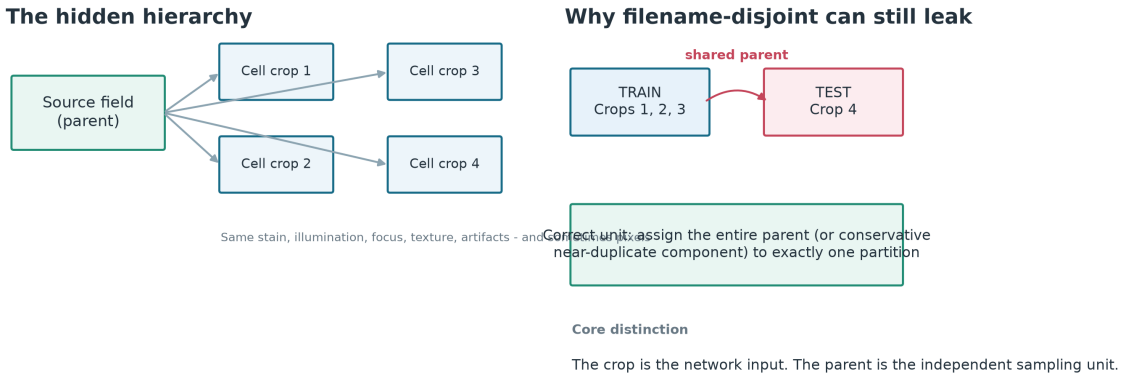
Do We Train on Test Cells?

The Complete CytoFair Project Record

Motivation, provenance recovery, leakage measurement, models, statistics, execution history, failures, verification, and reusable artifacts



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Frozen evidence through 7 September 2026
Comprehensive report compiled 13 September 2026 (Asia/Kolkata)

Document status and scope

This is the long-form technical record of the CytoFair project. It is intentionally broader than the six-page conference paper. It explains why the project was started, how every analysis was designed, how the code represents the data, what was measured, what worked, what failed, what was changed during execution, what remains unverified, and what artifacts can be reused. The report is written from the repository state and frozen result files; it does not add a new experiment.

Evidence boundary

The quantitative claims in this document come from committed CSV, JSON, Parquet, source, and report artifacts in the repository. The experiment inventory contains 120 SIPaKMeD classification runs, 30 parent-identity probe runs, and 175 Cx22 segmentation runs: 325 completed neural-network runs in total. Structural analyses are additional deterministic or Monte Carlo computations, not neural runs. No post-hoc classifier confusion matrix or qualitative model prediction is invented, because the frozen classifier JSON files retain aggregate metrics and epoch histories but not per-sample predictions or checkpoints.

How to interpret "leakage" in this report

Structural non-independence, literal pixel overlap, and a learnable parent fingerprint are demonstrated. These facts establish that a shortcut is available. They do *not* prove that every cervical-classification model causally uses that shortcut, and the multiplicity-corrected classifier contrasts do not establish a universal accuracy-inflation effect. This distinction is maintained throughout the report.

Abstract

Cervical-cell datasets are frequently released as crops, even when several crops came from one microscopic field or source image. A random crop split can therefore place sibling crops in both training and test sets, making file-level disjointness look like biological or acquisition-level independence. CytoFair recovers the hidden crop-to-parent hierarchy, validates it with fail-closed checks, measures structural and literal overlap, constructs parent- and component-disjoint splits, probes whether source identity remains learnable after transformations, and re-scores classification and segmentation models under matched protocols. The audit recovered all 4,049 SIPaKMeD crops, 1,320 Cx22 regions, 11,534 CRIC cells, and 917 Herlev cells. Under random splitting, 92.62% of SIPaKMeD test crops, 85.23% of Cx22 test regions, and 99.97% of CRIC test cells had a parent in training, whereas Herlev correctly remained at zero. More than half of SIPaKMeD test crops shared literal source pixels with a training crop. A 424-way parent probe reached 25.73% accuracy against 0.2358% chance. All six SIPaKMeD classifier P1–P2 point estimates and all seven Cx22 segmentation estimates were positive, but no confirmatory contrast survived its preregistered Benjamini–Hochberg family. The project therefore supports a strong benchmark-construction conclusion—evaluate at the independent parent unit—while retaining a narrower and appropriately uncertain model-performance conclusion.

Contents

Document status and scope i

1 Executive overview 1

1.1 The entire project in one page . . . 1

1.2 Headline findings, without spin . . . 2

2 Motivation and problem formulation 3

2.1 Why cervical cytology makes independence important . . . 3

2.2 Observation, parent, and evaluation unit . . . 3

2.3 Three different claims that must not be conflated . . . 4

2.4 Relationship to leakage and duplicate audits . . . 4

3 Preregistration and analysis governance 5

3.1 Why the plan was committed first . . . 5

3.2 Registered endpoints . . . 5

3.3 Amendments: what changed and why . . . 5

4 Datasets and the hidden hierarchy 7

4.1 Overview . . . 7

4.2 SIPaKMeD . . . 8

4.3 Cx22 and CCEDD . . . 9

4.4 CRIC . . . 9

4.5 Herlev as a structural negative control . . . 9

4.6 Data-use constraints . . . 9

5 System architecture and fail-closed philosophy 11

5.1 Pipeline . . . 11

5.2 Core artifact schemas . . . 12

5.3 Fail-closed algorithm . . . 12

6 Exact provenance recovery 13

6.1 SIPaKMeD Branch A: filename hierarchy . . . 13
6.2 SIPaKMeD Branch B: independent content matching . . . 13
6.3 Why the auxiliary acceptance flag is false for many correct rows . . . 14
6.4 Cx22, CRIC, and Herlev recovery . . . 14
6.5 Verification outcome . . . 15

7 Split protocols and structural leakage 16
7.1 Why four protocols are needed . . . 16
7.2 Analytic straddle probability . . . 17
7.3 Structural endpoints . . . 17
7.4 Interpretation . . . 18

8 Literal overlap and conservative duplicate grouping 20
8.1 Literal source-pixel overlap . . . 20
8.2 Perceptual hashes and P3 . . . 21

9 The parent-identity mechanism probe 22
9.1 Motivation . . . 22
9.2 Balanced task construction . . . 22
9.3 Image transformations and their motivations . . . 22
9.3.1 RGB . . . 22
9.3.2 Grayscale . . . 22
9.3.3 LAB stain normalization . . . 23
9.3.4 Background-only: Otsu, morphology, and Telea inpainting . . . 23
9.3.5 High-pass . . . 23
9.4 Shuffled-label negative control . . . 24
9.5 Results . . . 25

10 SIPaKMeD classification rescore 27
10.1 Why re-run models rather than quote published scores? . . . 27
10.2 Inputs, augmentation, and optimization . . . 27
10.3 Metric definitions . . . 28
10.4 Aggregate scores . . . 28
10.5 Matched contrasts, uncertainty, and the ordering failure . . . 29
10.6 What the classification rescore does and does not show . . . 31
10.7 Training behavior and compute . . . 32

11 Cx22 segmentation rescore 33
11.1 Why add a second task and dataset? . . . 33

11.2 Targets and models . . . 33
11.3 Deterministic input construction . . . 34
11.4 Semantic metrics . . . 34
11.5 Instance matching metric . . . 34
11.6 Semantic results . . . 35
11.7 Instance results . . . 36
11.8 Inference and rank stability . . . 37
11.9 Training failures, recovery, and runtime . . . 38

12 Statistical analysis in full **40**
12.1 The experimental unit is the seed-paired run . . . 40
12.2 Test-selection rule . . . 40
12.3 Bootstrap effect intervals . . . 40
12.4 Multiplicity families . . . 41
12.5 Structural sampling error . . . 41
12.6 Why a non-significant result is not “no effect” . . . 41

13 Execution lifecycle, failures, and demotivations **42**
13.1 Run inventory . . . 42
13.2 The first classification matrix was invalid . . . 42
13.3 MAnet numerical failure . . . 43
13.4 The host bottleneck . . . 43
13.5 Research stop and evidence freeze . . . 43

14 Verification against the acceptance contract **44**
14.1 Gate-by-gate status . . . 44
14.2 Why the classification ordering gate fails . . . 45
14.3 Release-readiness boundary . . . 45

15 Interpretation, limitations, and practical guidance **46**
15.1 What the evidence supports . . . 46
15.2 Two hard visual structures in Cx22 . . . 46
15.3 Real limitations . . . 46
15.4 Recommended evaluation rule . . . 47
15.5 What to say in a paper . . . 47

16 Reproduction and release engineering **48**
16.1 From owner downloads to structural Table 1 . . . 48
16.2 Clean-clone attestation . . . 49

16.3 Pinned software and result metadata . . . 49
16.4 Atomicity and resumability . . . 49
16.5 Human-readable and machine-readable layers . . . 49

17 Reusable outputs and future work 50
17.1 What exists now . . . 50
17.2 Two publication units . . . 50
17.3 Projects unblocked . . . 50
17.4 Pre-specified V2 in fish imagery . . . 51
17.5 Adoption goal . . . 51

18 Conclusion 52

A Complete SIPaKMeD classifier run ledger 53

B Complete parent-probe run ledger 58

C Complete Cx22 model run ledger 60

D Owner data files and integrity hashes 65

E Released split hashes 67

F Git decision chronology 68

G Exact command cookbook 70
G.1 Local checks . . . 70
G.2 Recover and verify structural evidence . . . 70
G.3 Launch the probe and classifier matrices . . . 70
G.4 Materialize and launch Cx22 . . . 71

H Artifact index and evidence status 72

I Complete derived-result ledgers 73
I.1 Structural and literal-overlap seeds . . . 73
I.2 Classifier contrasts . . . 75
I.3 Cx22 contrasts . . . 76
I.4 Rank stability . . . 77

J Excluded attempt ledger 78

K Reading guide for future maintainers 80

List of Figures

1.1 The project lifecycle. Blue events register decisions, green events record completed evidence, gold events document execution-driven amendments, red records a numerical failure, and purple marks this report. 2

2.1 The hidden hierarchy and the central failure mode. Several file-disjoint crops can share a parent field, source-specific nuisance factors, or literal source pixels. 4

4.1 Dataset scale and mean parent multiplicity. The log-scale left panel prevents CRIC's 11,534 observations from visually erasing the smaller datasets. 7

4.2 Complete parent-size distributions recovered from the Parquet maps. High multiplicity drives the probability that random crop splitting places a parent in both arms. 8

4.3 One released Cx22 Pair target decoded from the MATLAB-v7.3 structures. Cytoplasm and nucleus unions define two non-exclusive semantic channels; individual cytoplasm masks define the instance task. The CCEDD RGB source is not redistributed in this report. 9

5.1 Reproducibility architecture. Deterministic provenance and split artifacts gate the stochastic GPU experiments, and every final result carries the hashes needed to reconnect it to its inputs. 11

6.1 Branch-B diagnostics. The minimum winning NCC is 0.999969. The 0.02 runner-up-margin rule is intentionally only a diagnostic: 3,195 crops pass it and 854 do not, yet all rows agree with Branch A and all boxes remain contained. 13

6.2 A real recovered-coordinate example for the parent with the largest crop count. Only rectangle geometry is displayed; source pixels are not redistributed. Overlapping rectangles make literal source-pixel sharing possible. 14

7.1 The four protocols. P1 asks what common crop splitting shares. P1-prime holds its test set but matches the P2 training count. P2 groups exact parents. P3 groups whole near-duplicate-linked parent components. 16

7.2 The analytic mechanism. A one-crop parent cannot straddle; a high-multiplicity parent almost certainly straddles under independent 80:20 crop allocation. 17

7.3 Analytic and observed leakage. P2 and P3 are omitted from the bars because both are exactly zero on all four datasets. 18

7.4 Seed-level structural results. Reporting the spread prevents one convenient split from defining the claim. 18

8.1 Literal overlap for every P1 seed. The larger test-crop fraction answers a different question from the pair fraction: whether each test crop has at least one overlapping training sibling. 20

8.2 Near-duplicate sensitivity audit. The thresholded candidates are used conservatively for grouping; they are not treated as ground-truth duplicate annotations. 21

9.1 Every preprocessing stage shown on an actual local Herlev cell. This is an implementation demonstration, not a probe result image: the reported parent-identification experiment used SIPaKMeD crops. Panels D–G make Otsu thresholding, mask combination, and morphological closing visible before Telea inpainting. 24

9.2 All prespecified image conditions remain far above chance. Error bars are sample SD across five seeds. 25

9.3 Probe learning curves reconstructed from every retained epoch history. The RGB signal rises smoothly across seeds; shuffled labels remain near chance. 25

10.1 All frozen SIPaKMeD accuracy observations. Lines connect the same architecture across protocols; points show individual seeds and bars show the mean. 29

10.2 Paired P1−P2 effects. Every mean favors the crop-level split, but none survives its preregistered multiplicity correction. The vertical line at zero marks no protocol effect. 30

10.3 Seed-level paired differences. This figure exposes the small effective sample size and the architecture-specific variability that an aggregate table can hide. 30

10.4 The registered ordering check. Cell values are the protocol means; the title records the gate outcome. Four of six architectures fail P3≤P2≤P1, so the acceptance contract correctly records one failed gate. 31

10.5 Classifier wall-clock summaries from the retained run JSON files. These are same-system measurements and should not be compared to timings reported on unrelated hardware. 32

11.1 Cx22 semantic macro-Dice by architecture and protocol. The official arm is visually separated because it is not a same-observation contrast. 35

11.2 Channel-level Dice makes the central difficulty visible: nuclei are consistently harder than cytoplasm. This is the segmentation analogue of a confusion-matrix diagnostic and uses retained aggregate channel measurements, not invented pixel predictions. 36

11.3 Cx22 matched-instance Dice. Both Mask R-CNN variants follow the complete registered P3≤P2≤P1 ordering. 37

11.4 Cx22 paired contrast overview. Positive values favor the crop-level protocol. The plot separates point-estimate consistency from multiplicity-adjusted evidence. . . . 38

11.5 Cx22 run-time distribution. Instance segmentation dominates the computational cost and has a long tail; the plot makes the operational bottleneck explicit. 39

14.1 Machine-readable acceptance status: 13 gates pass, one fails, and three remain pending. A pending release gate is not painted green simply because the numerical experiments are complete. 44

16.1 The intended clean-clone path. Each boundary writes an auditable artifact and performs validation before the next stage begins. 48

List of Tables

1.1 Headline conclusions and their correct strength. 2

3.1 Registered amendments and their evidential status. 6

4.1 Audited datasets and independent units. 7

4.2 Independently reconstructed SIPaKMeD class totals in release order. 8

6.1 Provenance gates. 15

7.1 Structural leakage. Stochastic values are mean ± sample SD over five seeds. 18

9.1 Parent-identity probe. Mean ± sample SD over five seeds. 25

10.1 SIPaKMeD classifier execution settings. 28

10.2 SIPaKMeD test accuracy (%), mean ± sample SD over five seeds. P1′ is numerically identical to P1 because the frozen stratified crop split already contained the registered 3,239 training and 810 test observations. 28

10.3 Primary SIPaKMeD P1−P2 contrasts. The six paired t-tests form one preregistered BH family. CI is the paired-bootstrap 95% interval for Cohen's d_z, not for the raw percentage-point mean. 29

11.1 Cx22 segmentation execution settings. 34

11.2 Cx22 semantic macro-Dice (%), mean ± sample SD over five seeds. The official column uses a different 400/100 sample and is descriptive, not a paired protocol contrast. 35

11.3 Cx22 matched-instance Dice (%), mean ± sample SD over five seeds. 36

11.4 Primary Cx22 P1−P2 contrasts. Values are mean ± SD percentage points; q is within the task-specific preregistered family. Effect is Cohen's d_z except DeepLabV3+, whose non-normal seed differences trigger rank-biserial effect size. 37

13.1 Frozen neural-run inventory and summed run time. 42

14.1 Acceptance contract with evidence. 44

17.1 Immediate technical reuse. These rows describe enabled work, not completed future experiments. 50

A.1 Complete SIPaKMeD classification run ledger. All values are final-epoch results. . 54

B.1 Complete parent-identity probe ledger. 59

C.1 Complete Cx22 run ledger. Primary is macro Dice for semantic models and matched-instance Dice for instance models; the ninth column is nucleus Dice or recall at Dice 0.70. 61

D.1 Declared owner-source files, exact byte sizes, checksums, and use notes. 66

E.1 Released SIPaKMeD split manifests and hashes. 67

F.1 Complete repository decision chronology through the frozen result set. 68

H.1 Principal repository artifacts. 72

I.1 Complete structural seed ledger. 73

I.2 Complete SIPaKMeD literal source-pixel overlap ledger. 74

I.3 Complete SIPaKMeD classifier contrast ledger. Secondary tests have no confirmatory BH q-value. 75

I.4 Complete Cx22 contrast ledger. Semantic and instance primary families were corrected separately by task. 76

I.5 Complete secondary P1-versus-P2 rank-stability ledger. 77

J.1 Complete excluded-attempt ledger. Metrics are shown only to demonstrate that exclusion was rule-based, not score-based. 79

1

Executive overview

1.1 The entire project in one page

The initial problem was simple to state: *when a benchmark calls every image file an independent example, is that file really the unit that was independently sampled?* In cervical cytology, the answer can be no. One microscope field can emit several cell crops. If those crops are split independently, their source field can appear on both sides of evaluation.

The project followed a staged answer:

1. define the independent parent unit for four public datasets;
2. recover exact crop-to-parent mappings, using independent recovery branches where the release permits them;
3. refuse downstream analysis unless agreement, geometry, counts, and uniqueness pass;
4. calculate how often random crop splitting separates siblings;
5. detect literal pixel overlap and conservative cross-parent near-duplicate components;
6. create four fixed protocols that isolate independence from training-set size;
7. ask whether parent identity is visually learnable after removing candidate cues;
8. re-run multiple classifier and segmentation families over five fixed seeds;
9. use paired inference, effect sizes, bootstrap intervals, and declared BH families;
10. preserve failed attempts and execution amendments rather than erasing them.

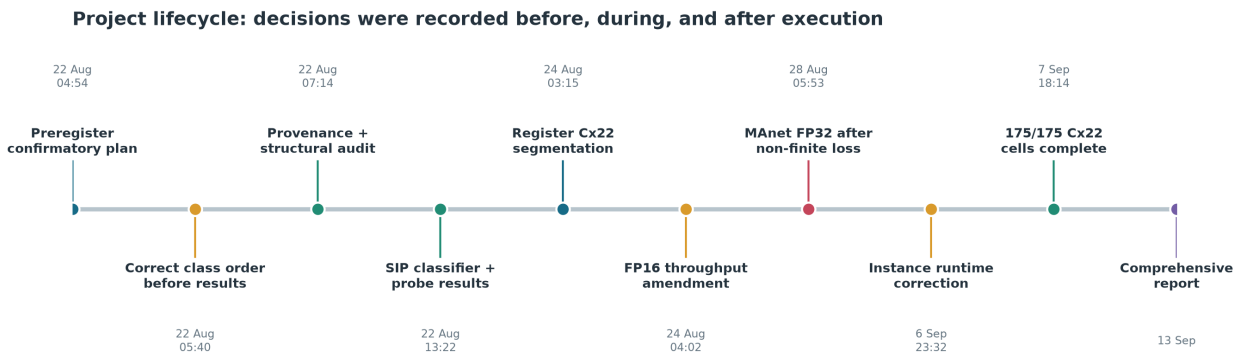


Figure 1.1: The project lifecycle. Blue events register decisions, green events record completed evidence, gold events document execution-driven amendments, red records a numerical failure, and purple marks this report.

1.2 Headline findings, without spin

Table 1.1: Headline conclusions and their correct strength.

Question	Answer	Evidence and boundary
Can provenance be recovered?	Yes	SIPaKMeD branches agree on 4,049/4,049 crops; all boxes are contained; all dataset and class totals match.
Does random crop splitting share parents?	Yes	Mean P1 test-parent overlap is 92.62% on SIPaKMeD, 85.23% on Cx22, and 99.97% on CRIC.
Does the negative control remain negative?	Yes	Herlev has one released cell per parent and returns exactly 0.00% structural leakage for all seeds.
Are literal pixels shared?	Yes, in SIPaKMeD P1	52.62% of test crops share at least one source pixel with a training crop; P2/P3 return zero.
Is source identity learnable?	Yes	A 424-way ResNet-18 reaches 25.73% versus 0.2358% chance; shuffled labels return 0.17% and cover chance.
Is P1 accuracy universally inflated?	Not established	All six SIPaKMeD and all seven Cx22 P1–P2 means are positive, but no contrast survives BH correction.
Does stronger grouping always lower score?	No	SIPaKMeD's registered P3 $\leq$ P2 $\leq$ P1 model ordering fails for four of six architectures. The failure is retained.
Is Cx22's official split defective?	No such claim	The authors' Multi-Train/Multi-Test parent sets are disjoint. The pooled stress test asks what happens under a different downstream reuse pattern.
Is the release fully ready?	Not yet	The clean-clone/virgin-host evidence, lab-paper crosswalk, and independent Figure-1 reader gate remain pending.

2

Motivation and problem formulation

2.1 Why cervical cytology makes independence important

Cervical cancer caused an estimated 660,000 new cases and 350,000 deaths worldwide in 2022 [29]. Cytology screening is a high-volume recognition problem: a small number of clinically important abnormal cells must be found among many plausible normal cells. Automation is attractive because it can help direct expert attention, but only if reported generalization represents genuinely unseen biological or acquisition units.

Modern cervical-cell classifiers report very high crop-level scores. DeepPap reported 98.3% on Herlev [30]; DeepCervix reported 99.14% on SIPaKMeD [20]; and a knowledge-distillation system reported 98.52% on SIPaKMeD [6]. These achievements motivated a basic audit question that is logically prior to choosing a stronger network: *what population unit does the test set represent?*

2.2 Observation, parent, and evaluation unit

Let x_i be a released image observation and let $g(x_i) = p_j$ map it to the acquisition unit that generated it. In this project:

- x_i is a SIPaKMeD cell crop and p_j is a cluster field;
- x_i is a Cx22 generated region and p_j is a CCEDD source image;
- x_i is a CRIC annotated cell and p_j is its source smear image;
- x_i is a Herlev single-cell image and is also its own parent.

The crucial idea is that the *model input* and the *independent sampling unit* need not be the same object. A network consumes x_i, but the statistical independence boundary may be p_j.

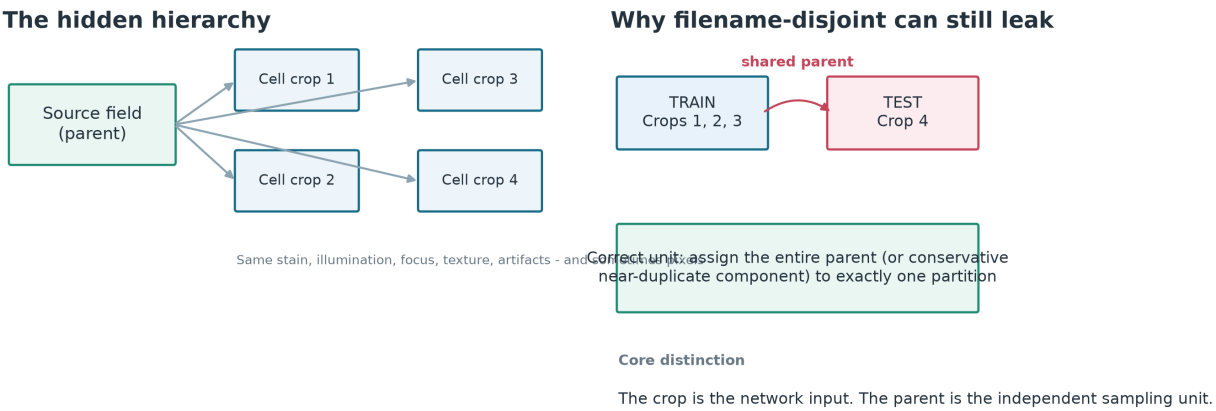


Figure 2.1: The hidden hierarchy and the central failure mode. Several file-disjoint crops can share a parent field, source-specific nuisance factors, or literal source pixels.

2.3 Three different claims that must not be conflated

Claim 1: structural non-independence

Some parents contribute observations to both train and test. This can be established exactly from provenance and split assignments. It requires no model.

Claim 2: shortcut availability

A model can identify the source parent from held-out sibling crops at greater-than-chance accuracy. This shows that parent information survives in the released input. It does not yet show that a disease or morphology classifier uses it.

Claim 3: downstream score inflation

A crop-level protocol produces a larger score than a parent-disjoint protocol under matched models, seeds, and sample sizes. This is an empirical paired comparison. It needs uncertainty, multiplicity correction, and careful interpretation.

The project obtains strong evidence for Claims 1 and 2. For Claim 3, point estimates are directionally consistent across every tested architecture, but the confirmatory families do not cross the multiplicity-adjusted threshold. The report therefore never upgrades "available shortcut" into "proven universal causal inflation."

2.4 Relationship to leakage and duplicate audits

Duplicate-aware evaluation has been studied in general vision: ciFAIR removed near-duplicate CIFAR test counterparts [1]; license-plate work re-evaluated models under duplicate-free splits [13]; and medical-image re-identification demonstrated that patient identity can be highly learnable [18]. Kapoor and Narayanan place these problems in a broader taxonomy of data leakage and reproducibility failures [11]. CytoFair's distinctive concern is an exact cervical-cytology source hierarchy: sibling crops can be non-independent without being duplicate files.

3

Preregistration and analysis governance

3.1 Why the plan was committed first

The original preregistration was committed at 04:54:10 IST on 22 August 2026 (commit `67e8ded`). The first result commit was recorded at 13:22:42 IST that day (commit `1c2da7f`). The plan therefore predates the committed results. This matters because a five-seed matrix contains many possible comparisons; fixing endpoints, seeds, exclusions, and multiplicity families before seeing scores reduces analyst degrees of freedom.

The fixed confirmatory seeds were $17, 29, 43, 71, 101$. Five additional robustness seeds were named but were not needed for the reported confirmatory summaries and were not substituted for any unfavorable seed. Every protocol-architecture cell had to contain at least five valid paired runs before inferential output could be emitted.

3.2 Registered endpoints

The structural endpoints were (i) fraction of parents straddling train and test and (ii) fraction of test observations whose parent appears in training. The mechanistic endpoint was 424-way parent-identification accuracy. The SIPaKMeD model endpoint was test accuracy; macro-F1 and balanced accuracy were concordance metrics. For Cx22 semantic segmentation, the primary endpoint was mean per-image cytoplasm/nucleus Dice. For Cx22 instance segmentation, the primary endpoint was per-image mean ground-truth-instance Dice after maximum-weight matching, with unmatched targets assigned zero.

3.3 Amendments: what changed and why

Table 3.1: Registered amendments and their evidential status.

Date	Amendment	Trigger	Why it did not select a favorable result
22 Aug	SIPaKMeD class order	The integrity gate exposed a transcription swap between Metaplastic and Dyskeratotic.	Corrected before any SIPaKMeD result; numeric target vector 126/108/238/271/223 did not change.
24 Aug	Cx22 segmentation extension	Structural Cx22 results motivated a separately scoped model rescore.	Registered before any Cx22 model was trained; official split retained as descriptive control.
24 Aug	FP16 execution	Full FP32 preflight left three A6000s under-used and made one U-Net epoch exceed 4.5 minutes.	Changed numerical execution before an epoch emitted a result; models, splits, seeds, endpoints, and epoch count stayed fixed.
28 Aug	MAnet in FP32	The first MAnet P1 seed-17 attempts produced non-finite losses at epochs 23 and 24.	No completed MAnet result existed. All MAnet cells restarted from their fixed seeds rather than selecting a pre-divergence checkpoint.
28 Aug	Deterministic CUDA workspace	Startup warnings showed deterministic matrix multiplication required `CUBLAS_WORKSPACE_CONFIG=:4096:8`.	Frozen before a corrected MAnet or instance epoch existed; applied to all remaining cells.
6 Sep	Instance runtime correction	Evaluation every epoch, CPU mask matching, and frequent network checkpoint writes caused severe under-utilization.	The registered endpoint was final epoch. Intermediate test inference was removed, exact binary intersections moved to GPU, and recovery checkpoints became five-epoch; final computation stayed identical.

A useful demotivation

The project began with the possibility of a large, architecture-general accuracy correction. The statistics did not support that broad claim. Instead of hiding this, the project narrowed its claim: the benchmark hierarchy is demonstrably non-independent, the source signal is demonstrably learnable, and parent grouping removes the measured sharing. The size of downstream score inflation remains model- and task-dependent.

4

Datasets and the hidden hierarchy

4.1 Overview

Table 4.1: Audited datasets and independent units.

Dataset	Observations	Parents	Mean/parent	Released task	Parent definition
SIPaKMeD	4,049	966	4.19	5-class classification	cluster field
Cx22	1,320	514	2.57	semantic/instance segmentation	CCEDD source image
CRIC	11,534	400	28.84	cytology classification resource	source smear image
Herlev	917	917	1.00	7-class single-cell classification	released cell image

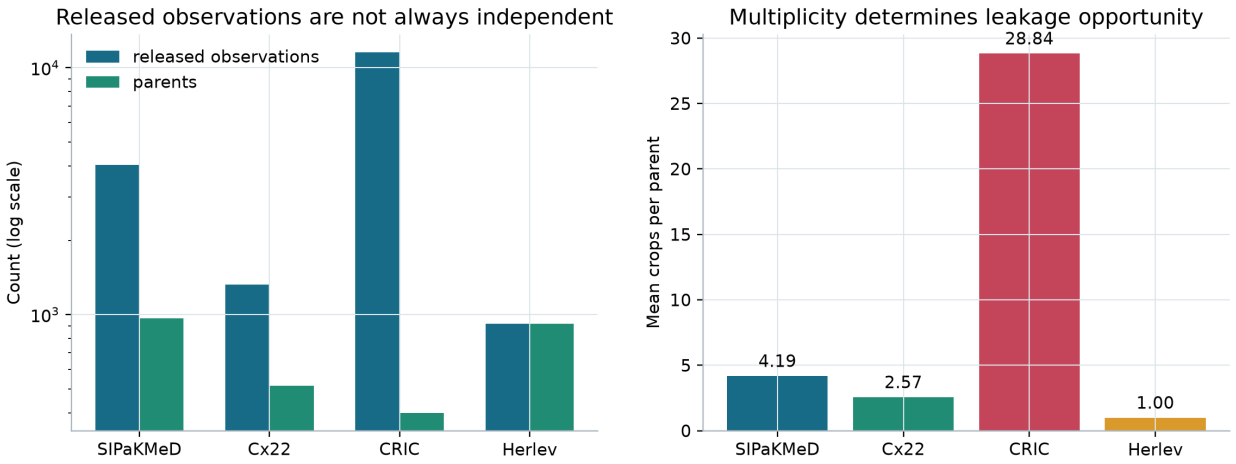


Figure 4.1: Dataset scale and mean parent multiplicity. The log-scale left panel prevents CRIC's 11,534 observations from visually erasing the smaller datasets.

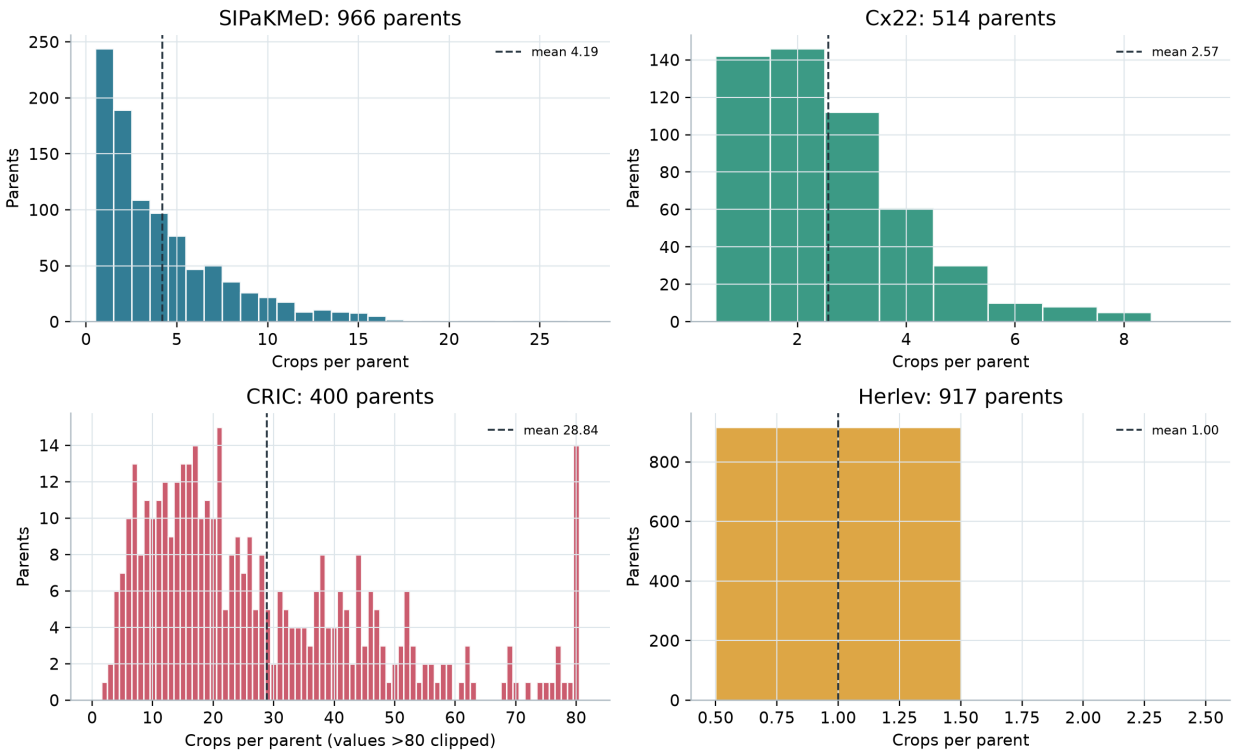


Figure 4.2: Complete parent-size distributions recovered from the Parquet maps. High multiplicity drives the probability that random crop splitting places a parent in both arms.

4.2 SIPaKMeD

SIPaKMeD contains 4,049 isolated cell images derived from 966 cluster fields [19]. The released hierarchy is unusually valuable for an audit: full fields and a `CROPPED` directory coexist, and crop filenames encode the field identifier. The project uses the cluster field—not the crop file—as the parent.

Table 4.2: Independently reconstructed SIPaKMeD class totals in release order.

Class	Crops	Parents	Crops/parent
Superficial–Intermediate	831	126	6.60
Parabasal	787	108	7.29
Koilocytotic	825	238	3.47
Metaplastic	793	271	2.93
Dyskeratotic	813	223	3.65
Total	4,049	966	4.19

The class order deserves explicit mention because the original planning text swapped the final two names even though its numerical vector was right. The integrity gate exposed that inconsistency, and Amendment 1 corrected it before results existed. This is an example of a check preventing a quiet metadata error.

4.3 Cx22 and CCEDD

Cx22 provides 1,320 generated regions and instance labels, drawing source images from CCEDD [15]. Three released subsets are involved: Multi-Train, Multi-Test, and Pair. The parent is the named CCEDD source image. Generator metadata provide source name and requested ROI coordinates; the authors' MATLAB generator defines how a source rectangle is clipped, resized, and pasted into a 512-by-512 white canvas.

The official Multi-Train and Multi-Test arms have zero shared parents. Pair is separately released and shares 55 parents with Multi-Test (Jaccard 0.126) and 233 with Multi-Train (Jaccard 0.475). The later model matrix pools all 1,320 observations only as a preregistered stress test of downstream reuse. It is not an accusation that the official authors used a leaking split.

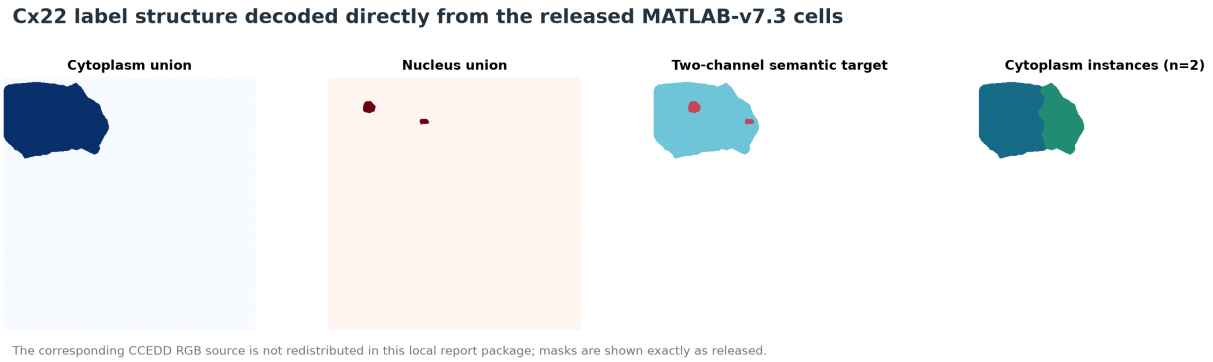


Figure 4.3: One released Cx22 Pair target decoded from the MATLAB-v7.3 structures. Cytoplasm and nucleus unions define two non-exclusive semantic channels; individual cytoplasm masks define the instance task. The CCEDD RGB source is not redistributed in this report.

4.4 CRIC

CRIC contains 11,534 classified cells associated with 400 source images [22]. The released classification table supplies source identity and cell coordinates, which makes source-level grouping deterministic. A reliable patient identifier is not present in the released table, so the project does not claim patient-disjoint evaluation. This matters: source-image grouping is stronger than crop grouping but can still be weaker than patient grouping.

4.5 Herlev as a structural negative control

Herlev contains 917 released single-cell images [10]. Under the project's declared data unit, each image is one observation and one parent. It therefore has no multi-observation parent structure and must return exactly zero parent straddling. If Herlev showed a nonzero structural effect, the implementation would be wrong. The control passed for all five seeds.

4.6 Data-use constraints

The repository publishes identifiers, coordinates, hashes, split assignments, code, and derived statistics. It does not redistribute raw images. Cx22/CCEDD material is restricted to non-commercial research/education in the declared source manifest, and CCEDD redistribution is prohibited. CRIC and the local Herlev repackaging are declared CC BY 4.0. SIPaKMeD's source

notes require citation and experimental use. The complete URL/checksum/terms ledger appears in table D.1.

5

System architecture and fail-closed philosophy

5.1 Pipeline

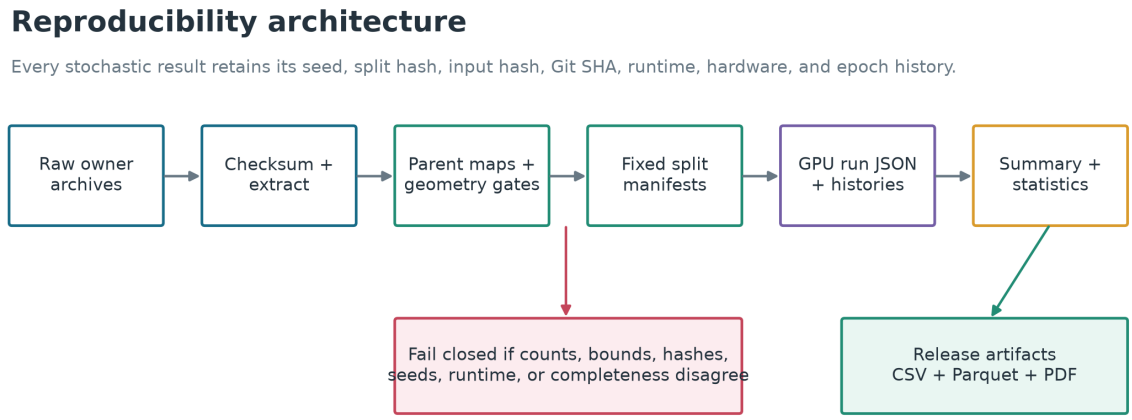


Figure 5.1: Reproducibility architecture. Deterministic provenance and split artifacts gate the stochastic GPU experiments, and every final result carries the hashes needed to reconnect it to its inputs.

The system deliberately separates three artifact layers:

1. **Canonical identity:** exact or released parent IDs used to construct splits.
2. **Independent audit:** image-matched parent IDs, boxes, confidence, runner-up margin, and an auxiliary diagnostic flag.
3. **Sensitivity grouping:** P3 connected-component IDs built from thresholded cross-parent near-duplicate candidates.

Keeping these layers separate prevents a lower-confidence image match from overwriting an exact filename identity. It also prevents a conservative pHash/dHash candidate from being narrated as a proven duplicate.

5.2 Core artifact schemas

The canonical SIPaKMeD map records `crop_id`, `parent_id`, `class_name`, paths, recovery method, and confidence. Branch B adds x, y, w, h, crop and parent dimensions, coarse and refined NCC, second-best confidence, margin, method, and diagnostic `accepted`. P3 adds `p3_group_id`. Split CSVs record `crop_id`, `parent_id`, `partition`, `protocol`, and `seed`.

Every neural result JSON records task, architecture, protocol/mode, seed, primary and secondary metrics, train/test counts, hyperparameters, exact split/input hashes, Git SHA, timestamps, elapsed time, visible GPU, platform and library versions, and full epoch history. This is more useful than a final number alone: it permits matrix completeness checks, runtime analysis, and learning-curve reconstruction without checkpoints.

5.3 Fail-closed algorithm

Listing 5.1: Conceptual fail-closed workflow.

```
for each released dataset:
    recover exact/released parent identity
    if an independent recovery branch exists:
        recover parent and bounding box without consulting branch A
        require >= 99% A/B agreement
    require one parent per observation
    require all boxes contained and all totals exact

construct P1, P1-prime, P2, and P3 for every fixed seed
measure structural and literal overlap
run probe/classifier/segmenter matrix with bounded GPUs

before summarizing:
    require every registered cell and seed exactly once
    require split hashes, runtime versions, and sample counts
    pair comparisons by seed
    attach an effect size and interval to every p-value
    apply BH over the entire declared family once

if any required evidence is absent:
    emit FAIL or PENDING; never infer PASS
```

6

Exact provenance recovery

6.1 SIPaKMeD Branch A: filename hierarchy

Branch A parses the released directory and filename convention. For example, `im_Dyskeratotic/CROPPED/001_03.bmp` maps to parent `Dyskeratotic/001`. This route is deterministic, fast, and used for the canonical split map. Its weakness is that a parsing bug could produce internally tidy but wrong identifiers. That is why Branch B was required.

6.2 SIPaKMeD Branch B: independent content matching

For crop x and candidate parent region r, normalized cross-correlation is

$$\mathrm{NCC}(x, r) = \frac{\sum_u (x_u - \bar{x})(r_u - \bar{r})}{\sqrt{\sum_u (x_u - \bar{x})^2} \sqrt{\sum_u (r_u - \bar{r})^2}}. \tag{6.1}$$

Branch B never uses Branch A's parent assignment in the query. It ranks every same-class parent at quarter resolution, refines the five best candidates at full resolution, and records the winning location as a bounding box. A multiscale fallback exists only for unresolved cases. Matching is bounded by class working sets so the complete parent image collection does not have to remain in memory.

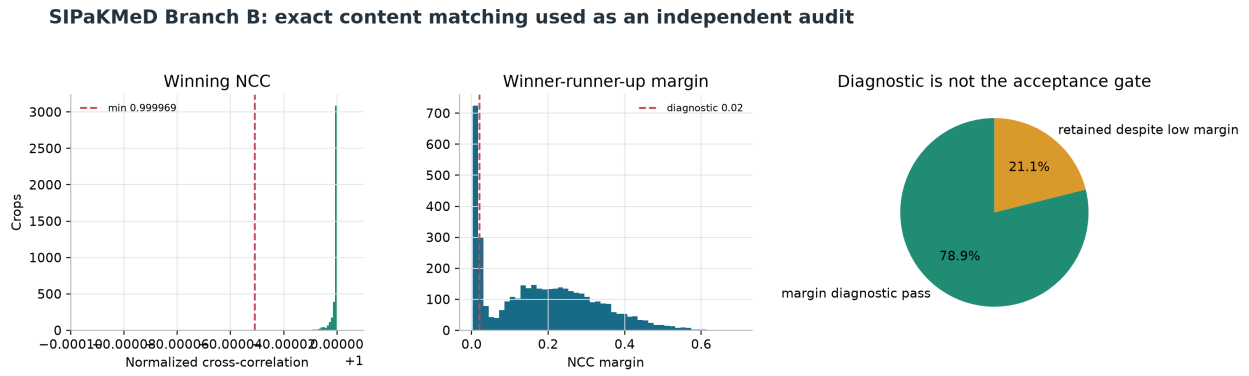


Figure 6.1: Branch-B diagnostics. The minimum winning NCC is 0.999969. The 0.02 runner-up-margin rule is intentionally only a diagnostic: 3,195 crops pass it and 854 do not, yet all rows agree with Branch A and all boxes remain contained.

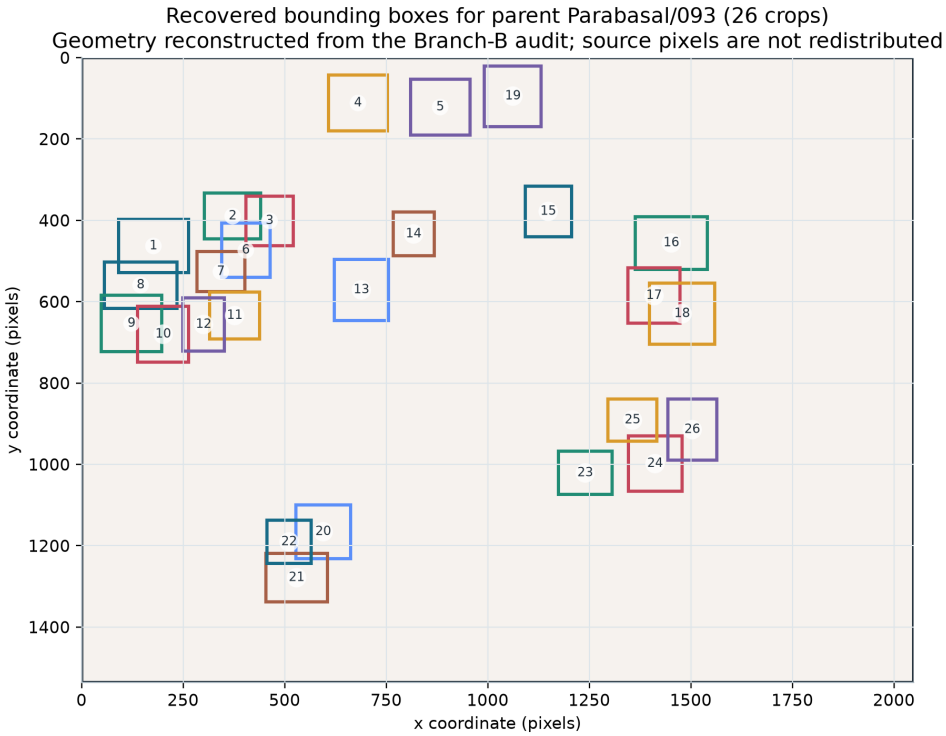


Figure 6.2: A real recovered-coordinate example for the parent with the largest crop count. Only rectangle geometry is displayed; source pixels are not redistributed. Overlapping rectangles make literal source-pixel sharing possible.

6.3 Why the auxiliary acceptance flag is false for many correct rows

The `accepted` field in Branch B encodes a conservative confidence-plus-margin diagnostic. It is not the registered provenance acceptance outcome. Many microscope fields contain visually repetitive background, so a correct near-perfect winner can have another near-perfect candidate. Substituting this diagnostic for the registered 99% cross-branch agreement rule would discard 854 correctly mapped crops after looking at an arbitrary margin. The project instead retains every row and exposes the ambiguity measure.

6.4 Cx22, CRIC, and Herlev recovery

Cx22 uses source names and ROI coordinates decoded from the authors' three MATLAB generator releases. Every source resolves against the declared 686-image CCEDD set, and rare edge-touching requested boxes are clipped exactly as the generator code specifies. The released Cx22 archives contain generator metadata and masks rather than a second independent RGB crop array, so an A/B image-content branch cannot be constructed from the public package alone. Applicable checks are source existence, exact metadata decoding, generator geometry, bounds, and subset overlap.

CRIC uses the released classifications table: source-image identity and coordinates directly define the map. Herlev assigns each released cell to itself. Neither route pretends to recover unavailable patient identity.

6.5 Verification outcome

Table 6.1: Provenance gates.

Dataset	Observations	Parents	Applied evidence	Outcome
SIPaKMeD	4,049	966	4,049/4,049 A/B agreement; 4,049 contained boxes; exact class totals	**PASS**
Cx22	1,320	514	generator names/ROIs; 686 sources resolve; 1,320 contained clipped boxes	**PASS**
CRIC	11,534	400	released source identities and coordinates; no patient-ID claim	**PASS**
Herlev	917	917	one released cell per parent	**PASS**

No observation was excluded. The five SIPaKMeD parent counts are exactly 126/108/238/271/223 in corrected release order. All crop counts sum to 4,049.

7

Split protocols and structural leakage

7.1 Why four protocols are needed

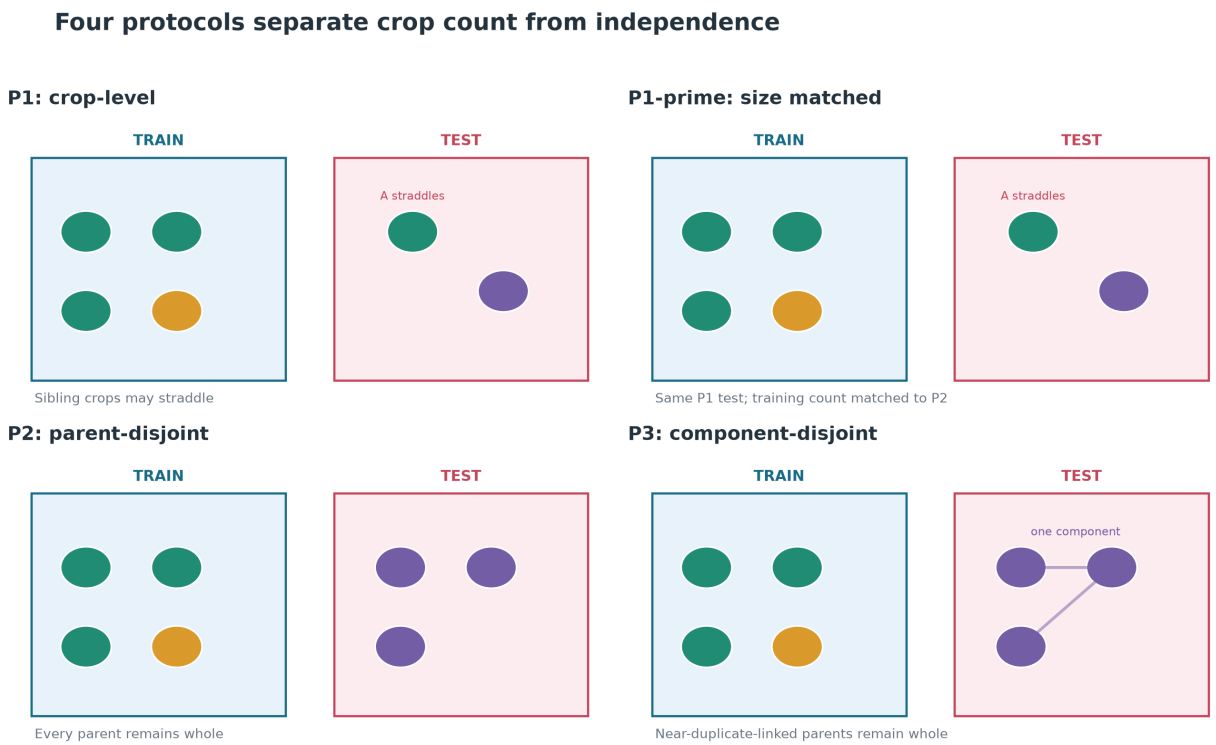


Figure 7.1: The four protocols. P1 asks what common crop splitting shares. P1-prime holds its test set but matches the P2 training count. P2 groups exact parents. P3 groups whole near-duplicate-linked parent components.

All splits use an 80:20 target and the five fixed seeds. P1 is class-stratified at crop level. P1-prime retains P1's test observations and downsamples only its training observations to exactly the P2 training count, as closely class-balanced as integer counts permit. P2 uses group-aware allocation by exact parent. P3 repeats parent-disjoint allocation after cross-parent candidate links are collapsed into connected components.

On the frozen SIPaKMeD environment, P2 contains 3,239 train and 810 test crops for every seed. P1 already has exactly those counts, so P1-prime is identical to P1. This is not a redundant

hidden comparison: it shows that the primary P1–P2 difference cannot be attributed to a larger P1 training set in this implementation.

7.2 Analytic straddle probability

For a parent with k crops independently assigned to training with probability 0.8 and test with probability 0.2, the only non-straddling outcomes are "all train" and "all test." Hence

$$P(\text{no straddle} \mid k) = 0.8^k + 0.2^k, \tag{7.1}$$

$$P(\text{straddle} \mid k) = 1 - 0.8^k - 0.2^k. \tag{7.2}$$

For a dataset with parent crop counts $k_1, \ldots, k_m$, the analytic expectation is

$$\bar{P}_{\text{straddle}} = \frac{1}{m} \sum_{j=1}^{m} \left(1 - 0.8^{k_j} - 0.2^{k_j}\right). \tag{7.3}$$

The exact class-stratified fixed-size splitter is not perfectly independent Bernoulli assignment. Therefore the analytic quantity is checked against a 10,000-allocation Monte Carlo interval generated with the actual count vector and allocation rule.

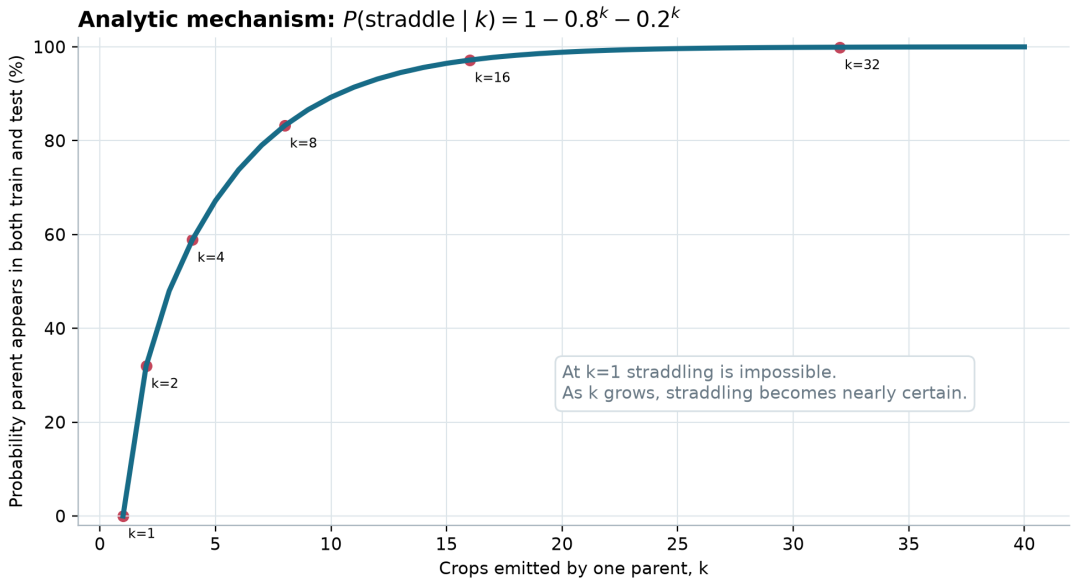


Figure 7.2: The analytic mechanism. A one-crop parent cannot straddle; a high-multiplicity parent almost certainly straddles under independent 80:20 crop allocation.

7.3 Structural endpoints

Let G_{tr} and G_{te} be training and test parent sets, and let T be the test observations. The parent-straddle fraction and leaked-test-observation fraction are

$$L_p = \frac{|G_{\text{tr}} \cap G_{\text{te}}|}{|G|}, \qquad L_c = \frac{1}{|T|} \sum_{x \in T} \mathbf{1}\{g(x) \in G_{\text{tr}}\}. \tag{7.4}$$

Table 7.1: Structural leakage. Stochastic values are mean ± sample SD over five seeds.

Dataset	Crops	Parents	Analytic P1 straddle	Observed P1 straddle	P1 test with train parent	P2/P3 straddle
SIPaKMeD	4,049	966	44.83%	45.63±1.04%	92.62±0.40%	0.00±0.00%
Cx22	1,320	514	33.93%	34.01±1.10%	85.23±3.51%	0.00±0.00%
CRIC	11,534	400	94.88%	94.35±1.10%	99.97±0.08%	0.00±0.00%
Herlev	917	917	0.00%	0.00±0.00%	0.00±0.00%	0.00±0.00%

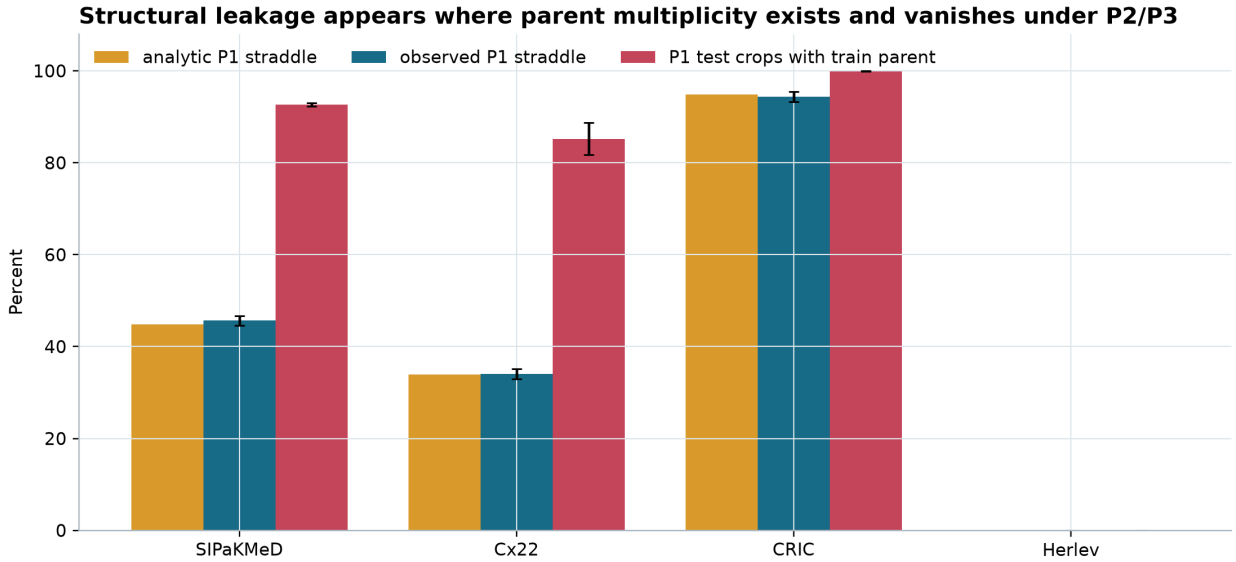


Figure 7.3: Analytic and observed leakage. P2 and P3 are omitted from the bars because both are exactly zero on all four datasets.

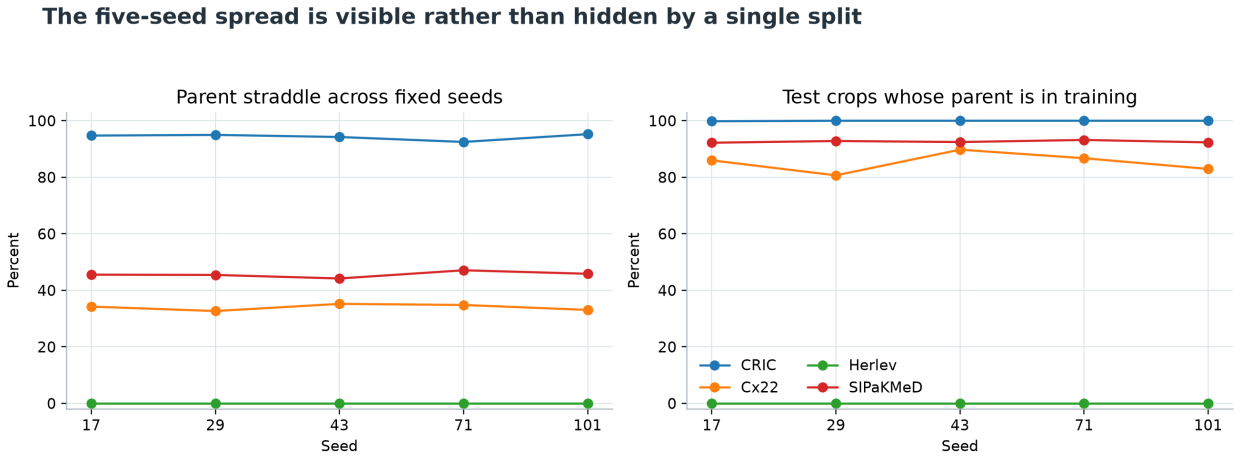


Figure 7.4: Seed-level structural results. Reporting the spread prevents one convenient split from defining the claim.

7.4 Interpretation

CRIC is the most extreme because its parents emit 28.84 crops on average: essentially every random-split test cell has a source image in training. SIPaKMeD and Cx22 have lower parent multiplicity, yet 92.62% and 85.23% of their test observations respectively still have a training sibling. Herlev establishes implementation specificity: the same code returns zero when the data

genuinely have one observation per parent.

The structural ordering P3 $\leq$ P2 $\leq$ P1 passes for all datasets. This statement concerns shared parents, not model scores. Later, the analogous model-score ordering fails for four of six SIPaKMeD classifiers; the two outcomes are logically compatible.

8

Literal overlap and conservative duplicate grouping

8.1 Literal source-pixel overlap

SIPaKMeD's recovered boxes permit a stronger question than shared parent identity. Let boxes B_i and B_j come from the same field. Their intersection is non-empty when

$$\max(x_i, x_j) < \min(x_i + w_i, x_j + w_j) \quad \text{and} \quad \max(y_i, y_j) < \min(y_i + h_i, y_j + h_j). \tag{8.1}$$

The intersection-over-union is

$$\mathrm{IoU}(B_i, B_j) = \frac{|B_i \cap B_j|}{|B_i \cup B_j|}. \tag{8.2}$$

Across 13,064 within-parent crop pairs, 2,185 share at least one pixel and the maximum IoU is 0.546. Under P1, 16.80±0.77% of cross-split sibling pairs overlap and 52.62±1.97% of test crops overlap at least one source pixel with a training crop. Both quantities are 0.00±0.00% under P2 and P3.

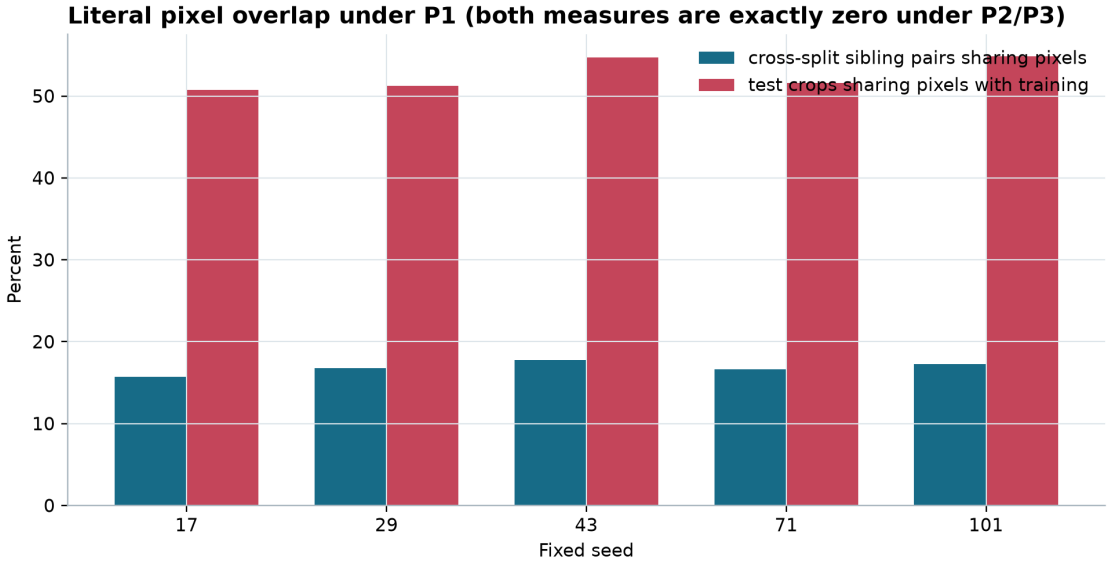


Figure 8.1: Literal overlap for every P1 seed. The larger test-crop fraction answers a different question from the pair fraction: whether each test crop has at least one overlapping training sibling.

8.2 Perceptual hashes and P3

For each crop the implementation computes a 64-bit perceptual hash from the median-thresholded low-frequency 8×8 DCT block and a 64-bit difference hash from adjacent comparisons in an 8×9 downsample. Two crops become a candidate pair when they are pixel-exact or when both Hamming distances are at most six. Cross-parent candidate pairs are joined by union-find; each connected component becomes one P3 group.

At the frozen thresholds, the audit found 201 cross-parent candidate pairs. They connect 170 parent IDs into 74 components, the largest containing eight parents, and reduce 966 parents to 870 P3 groups. No pair is pixel-exact. A synthetic calibration detected all three injected transformations and rejected all 100 deterministic unrelated-parent negatives, yielding precision 1.00 and recall 1.00 on that calibration set.

P3 is a conservative sensitivity grouping, not a claim that every candidate is a duplicate

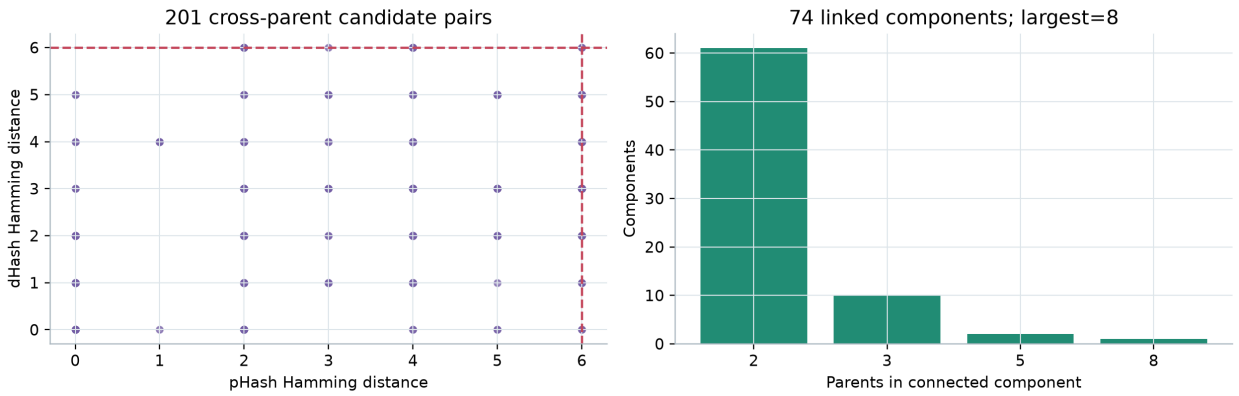


Figure 8.2: Near-duplicate sensitivity audit. The thresholded candidates are used conservatively for grouping; they are not treated as ground-truth duplicate annotations.

What the duplicate experiment did not do

The broader planning idea mentioned a three-detector majority vote involving learned features. The frozen release implements only the pHash+dHash consensus audit and its synthetic calibration. Therefore the report does not claim a CNN/domain-encoder duplicate ensemble, human-confirmed duplicate truth, or general detector precision outside the calibration set.

9

The parent-identity mechanism probe

9.1 Motivation

Structural overlap says that siblings are present. The parent probe asks whether their shared source is visually recoverable after cellular content and color cues are altered. A positive answer makes the shortcut concrete: a learner can recognize the parent field from a different crop. It still does not prove that a morphology classifier uses that signal.

9.2 Balanced task construction

Parents with at least four crops are eligible. The minimum eligible size allows two training and two test crops per parent. To make chance exact, each eligible parent contributes the same number of training and test crops. The final task contains 424 parents, 848 training crops, and 848 test crops, so chance is

$$P_{\text{chance}} = \frac{1}{424} = 0.00235849 = 0.235849\%. \tag{9.1}$$

A ResNet-18 [7] initialized from ImageNet replaces its final layer with a 424-way linear classifier. It trains for 15 epochs with AdamW [16], learning rate 10^{-4}, weight decay 10^{-4}, batch size 64, and fixed seeds. Training augmentation is a random resized crop with scale 0.85–1.0 plus horizontal and vertical flips; testing resizes to 224×224. ImageNet normalization follows every image-mode transform.

9.3 Image transformations and their motivations

9.3.1 RGB

RGB is the maximal-information condition. It permits morphology, stain, illumination, background, focus, acquisition artifacts, and any other visible cue.

9.3.2 Grayscale

Grayscale removes hue and saturation while retaining luminance, edges, morphology, focus, and texture. Persistence here means identity is not solely a stain-color phenomenon.

9.3.3 LAB stain normalization

RGB is converted to CIE LAB. For channel vector z, source mean μ_s, source standard deviation σ_s, and fixed target statistics $\mu_t = (190, 140, 135)$ and $\sigma_t = (35, 14, 14)$, the implemented transform is

$$z' = \text{clip}\left((z - \mu_s) \odot \frac{\sigma_t}{\max(\sigma_s, 1)} + \mu_t, 0, 255\right). \tag{9.2}$$

This is a fixed-statistics color-transfer normalization based on the logic of Reinhard et al. [21]. Persistence means simple first- and second-moment color normalization does not erase the source fingerprint.

9.3.4 Background-only: Otsu, morphology, and Telea inpainting

This condition deliberately removes the likely cell foreground. The implementation converts RGB to HSV and grayscale, then independently obtains Otsu thresholds [17] on saturation S and luminance Y. The preliminary foreground mask is

$$M_0(u) = \mathbf{1}\{S(u) > t_S \ \vee \ Y(u) < t_Y\}. \tag{9.3}$$

A 5×5 morphological closing fills small holes and connects narrow gaps:

$$M = (M_0 \oplus K) \ominus K. \tag{9.4}$$

Finally, the marked region is filled from its boundary with Telea's fast-marching inpainting method using radius 5 [27]. The result keeps background statistics while suppressing the cell body and nucleus. A successful classifier in this condition shows that background alone carries considerable parent information.

9.3.5 High-pass

For Gaussian blur $G_{\sigma=3} * I$, the residual is

$$I_{\text{hp}} = \text{clip}\left(I - G_{\sigma=3} * I + 128, 0, 255\right). \tag{9.5}$$

This suppresses low-frequency illumination and emphasizes local texture, edges, compression, and focus structure.

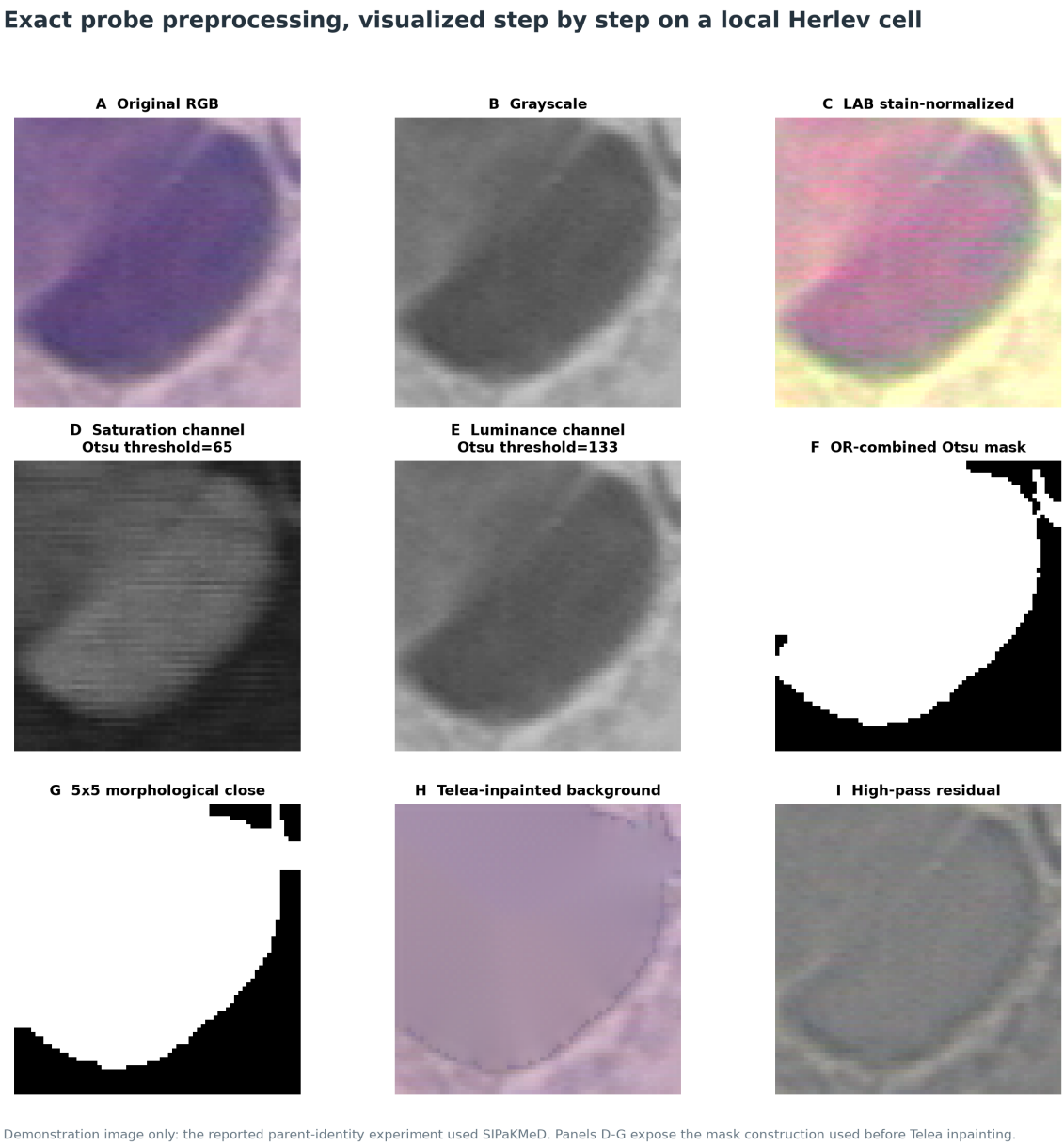


Figure 9.1: Every preprocessing stage shown on an actual local Herlev cell. This is an implementation demonstration, not a probe result image: the reported parent-identification experiment used SIPaKMeD crops. Panels D–G make Otsu thresholding, mask combination, and morphological closing visible before Telea inpainting.

9.4 Shuffled-label negative control

Only the training labels are permuted, using the same images, optimizer, epochs, seeds, and task size. The five-seed 95% interval must cover the exact chance baseline. Otherwise the probe would be rejected because a bookkeeping or imbalance artifact could explain apparent identity learning.

9.5 Results

Table 9.1: Parent-identity probe. Mean ± sample SD over five seeds.

Input condition	Accuracy	Multiple of chance
RGB	25.73±0.66%	109.1×
Grayscale	13.80±1.12%	58.5×
Stain-normalized	17.31±0.64%	73.4×
Background-only	18.68±1.25%	79.2×
High-pass	16.27±0.60%	69.0×
Shuffled RGB labels	0.17±0.24%	0.7×

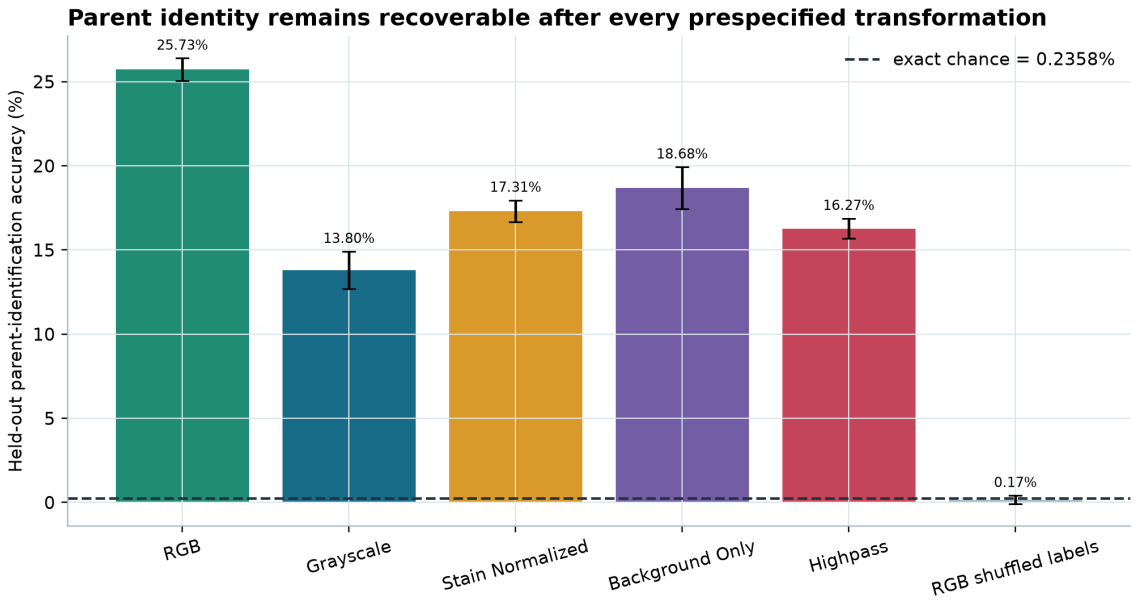


Figure 9.2: All prespecified image conditions remain far above chance. Error bars are sample SD across five seeds.

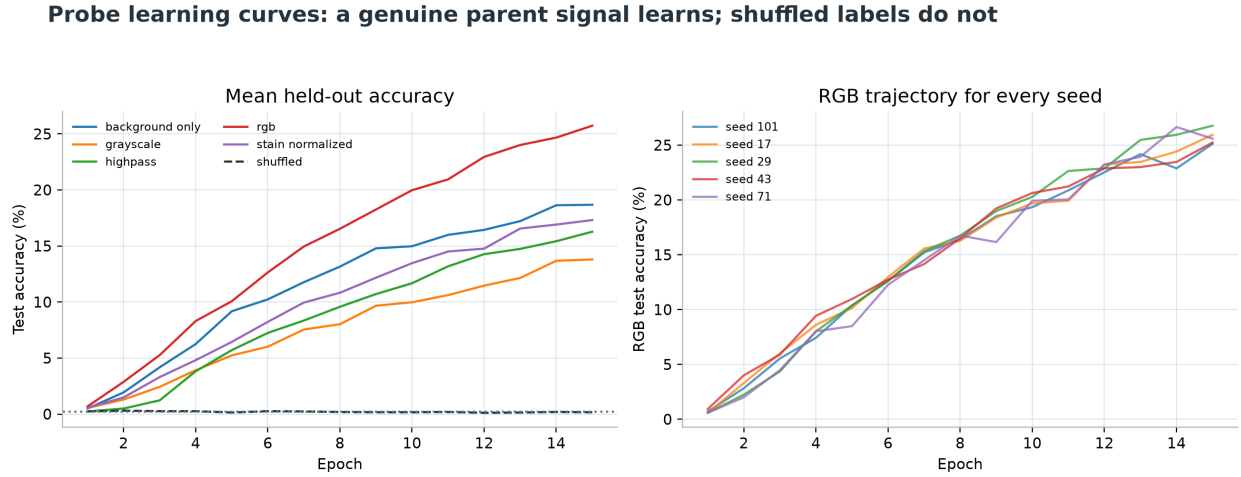


Figure 9.3: Probe learning curves reconstructed from every retained epoch history. The RGB signal rises smoothly across seeds; shuffled labels remain near chance.

The shuffled mean is 0.1651%, SD 0.2445 percentage points, with a five-seed interval from −0.1385% to 0.4687%; the exact 0.2358% chance value lies inside. A negative lower confidence limit is an ordinary symmetric small-sample interval artifact and does not imply negative accuracy.

Background-only accuracy is higher than grayscale and stain-normalized accuracy. That does not mean background is the only cue: RGB remains highest, and transformations alter multiple image properties at once. The careful conclusion is that parent information is distributed across color, low- and high-frequency structure, and background.

10

SIPaKMeD classification rescore

10.1 Why re-run models rather than quote published scores?

Published numbers are entangled with architectures, preprocessing, augmentations, and—most importantly—different splits. Quoting them would not isolate the quantity of interest. The rescore therefore trains six representative image classifiers under the same code path, preprocessing, optimizer family, epoch budget, seeds, and four fixed split protocols. The only intended change across protocols is the independence constraint and, for the explicitly size-matched control, the number of retained training crops.

The six families cover different inductive biases and capacities: ResNet-18 and ResNet-50 [7], DenseNet-121 [9], MobileNet-V2 [24], EfficientNet-B0 [26], and ViT-B/16 [4]. Each ImageNet-pretrained head is replaced with a five-way linear classifier. Re-running all six also prevents the conclusion from depending on one especially favorable architecture.

10.2 Inputs, augmentation, and optimization

Every crop is decoded as RGB, converted to a tensor, and normalized by the ImageNet channel statistics. Training augmentation consists of a random resized crop to 224×224 with scale in $[0.8, 1.0]$, a horizontal flip, and a vertical flip. Test preprocessing is a deterministic resize to 224×224. The code records the fully resolved split hash and configuration beside every result.

All classifiers use AdamW [16], cross-entropy loss, and a fixed learning rate. The retained endpoint is the final epoch; there is no separate validation split or post-hoc best-epoch selection. A run is valid only when the stored identifiers match its split, the final file is complete, and all reported quantities are finite. Table 10.1 gives the execution contract.

Table 10.1: SIPaKMeD classifier execution settings.

Item	Setting
Architectures	ResNet-18/50, DenseNet-121, MobileNet-V2, EfficientNet-B0, ViT-B/16
Initialization	ImageNet pretrained
Input	RGB, 224×224
Loss	Five-class cross entropy
Optimizer	AdamW
Learning rate	10^{-4}, fixed
Weight decay	10^{-4}
Batch size	64
Epochs	20, final-epoch endpoint
Seeds	17, 29, 43, 71, 101
Primary metric	Test accuracy
Hardware	NVIDIA RTX A6000; at most three concurrent GPU jobs

10.3 Metric definitions

For true class y_i and prediction $\hat{y}_i$, accuracy is

$$\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}. \tag{10.1}$$

For class c, with precision P_c and recall R_c,

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c}, \qquad F_{1,\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c}. \tag{10.2}$$

Balanced accuracy is the unweighted mean class recall,

$$\mathrm{BAcc} = \frac{1}{C}\sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c}. \tag{10.3}$$

The preregistered primary endpoint is accuracy. Macro-F1 and balanced accuracy check that the direction is not a class-frequency artifact; they are not used to shop for a smaller p-value.

10.4 Aggregate scores

Table 10.2: SIPaKMeD test accuracy (%), mean ± sample SD over five seeds. P1′ is numerically identical to P1 because the frozen stratified crop split already contained the registered 3,239 training and 810 test observations.

Architecture	P1	P1′	P2	P3
DenseNet-121	98.12±0.28	98.12±0.28	96.72±0.69	97.38±0.50
EfficientNet-B0	97.78±0.49	97.78±0.49	96.69±1.49	97.41±0.71
MobileNet-V2	97.41±0.45	97.41±0.45	95.16±1.50	96.20±0.93
ResNet-18	97.19±0.59	97.19±0.59	96.00±0.89	96.35±1.28
ResNet-50	97.51±0.47	97.51±0.47	96.79±0.42	96.27±0.53
ViT-B/16	97.14±0.95	97.14±0.95	95.56±0.84	95.53±1.68

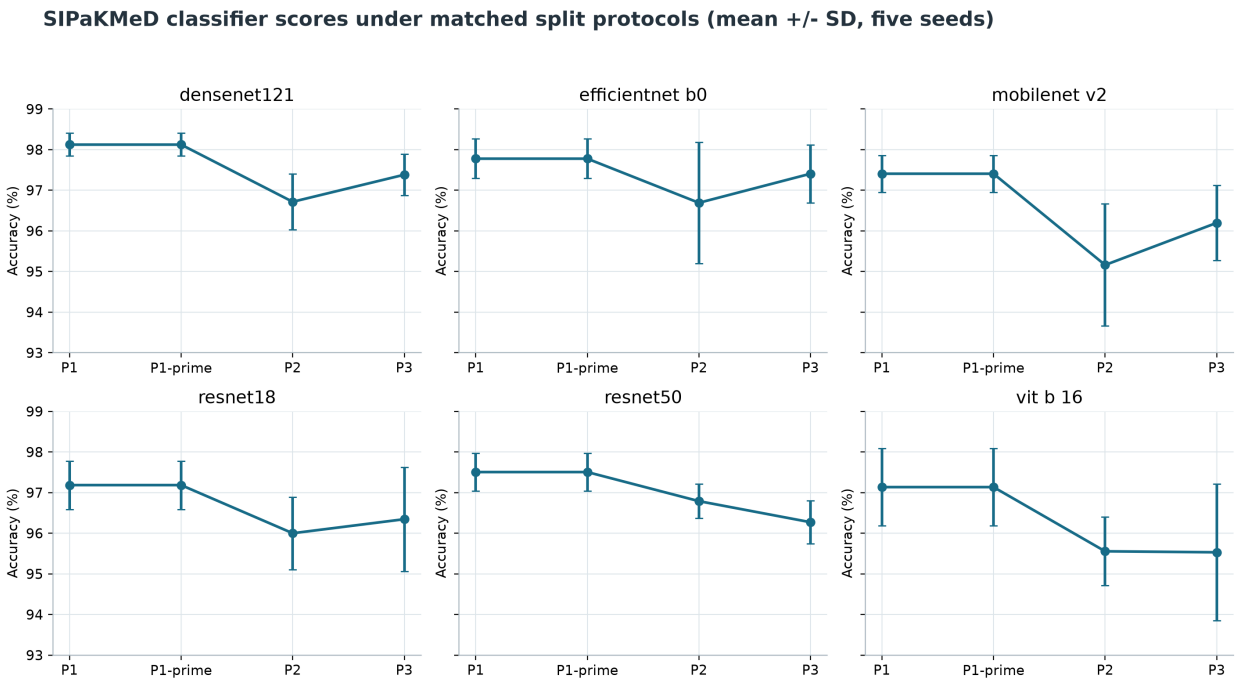


Figure 10.1: All frozen SIPaKMeD accuracy observations. Lines connect the same architecture across protocols; points show individual seeds and bars show the mean.

The top score under P1 is DenseNet-121 at 98.12±0.28%. Under parent-disjoint P2, DenseNet-121 remains highest at 96.72±0.69%, narrowly followed by ResNet-50. The key scientific observation is not which backbone wins; it is that all six paired P1−P2 mean differences are positive.

10.5 Matched contrasts, uncertainty, and the ordering failure

For architecture a and seed s, the primary paired difference is

$$d_{a,s} = 100\,(\mathrm{Acc}_{a,s,\mathrm{P1}} - \mathrm{Acc}_{a,s,\mathrm{P2}}) \tag{10.4}$$

in percentage points. The paired standardized effect is Cohen's

$$d_z = \frac{\bar{d}}{s_d}, \tag{10.5}$$

where s_d is the sample SD of the five seed-paired differences. Reported effect-size confidence intervals are seed-paired bootstrap intervals, so a bootstrap resample always keeps the two protocol results from a seed together. The test is a paired t-test because every registered five-difference vector passed its Shapiro–Wilk screen; the switch to Wilcoxon was reserved for a failed screen.

Table 10.3: Primary SIPaKMeD P1−P2 contrasts. The six paired t-tests form one preregistered BH family. CI is the paired-bootstrap 95% interval for Cohen's d_z, not for the raw percentage-point mean.

Architecture	Mean±SD (pp)	95% CI for d_z	p	BH q	d_z
DenseNet-121	1.41±0.90	[0.985, 4.747]	0.0251	0.1109	1.561
EfficientNet-B0	1.09±1.85	[−0.316, 4.668]	0.2591	0.2591	0.588
MobileNet-V2	2.25±1.63	[0.809, 6.686]	0.0370	0.1109	1.377
ResNet-18	1.19±1.28	[0.442, 2.365]	0.1081	0.1511	0.922
ResNet-50	0.72±0.83	[0.085, 4.025]	0.1260	0.1511	0.863
ViT-B/16	1.58±1.70	[0.057, 4.950]	0.1059	0.1511	0.931

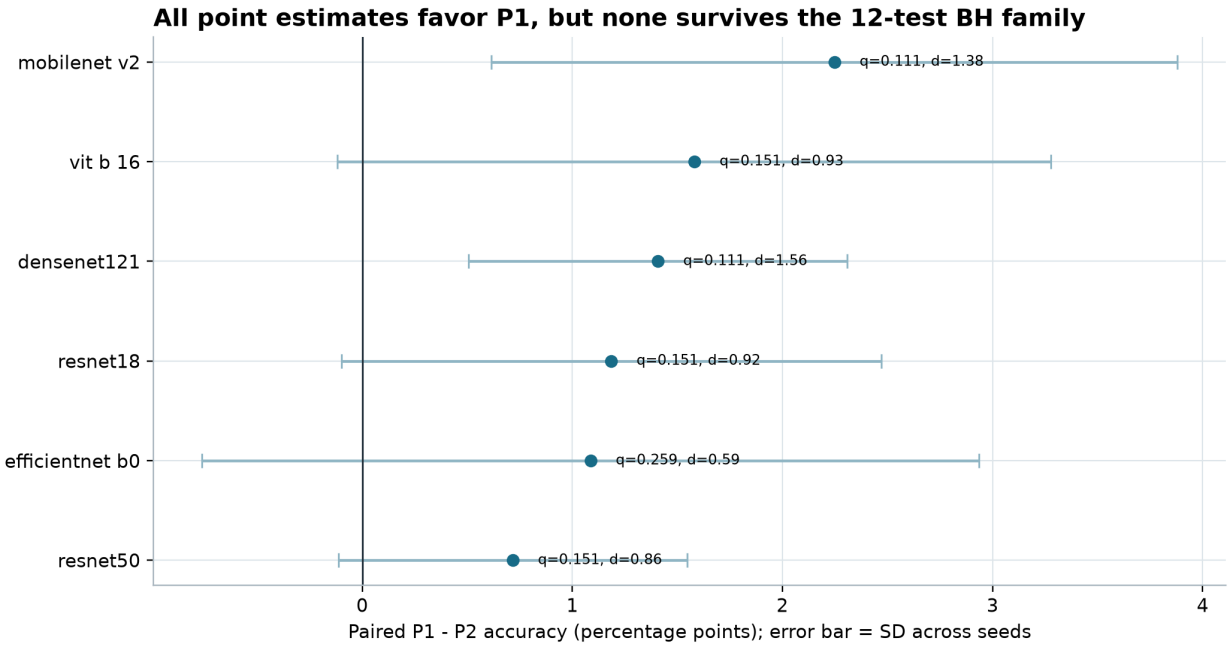


Figure 10.2: Paired P1−P2 effects. Every mean favors the crop-level split, but none survives its preregistered multiplicity correction. The vertical line at zero marks no protocol effect.

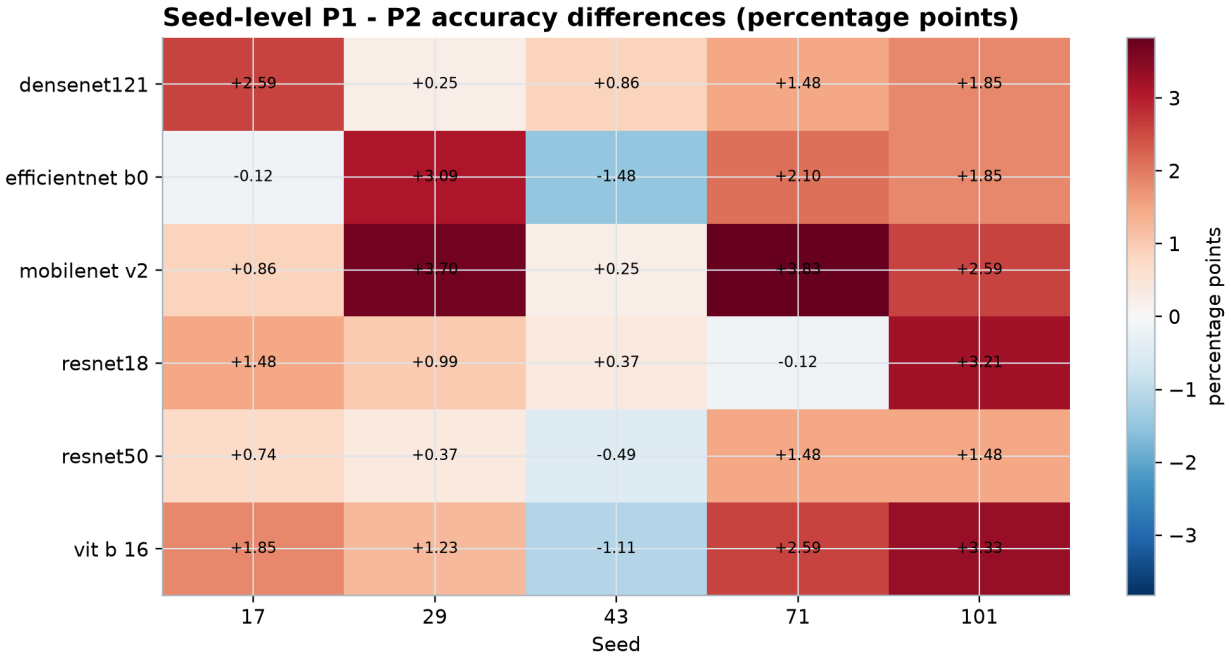


Figure 10.3: Seed-level paired differences. This figure exposes the small effective sample size and the architecture-specific variability that an aggregate table can hide.

The registered monotonicity expectation was P3≤P2≤P1. Only ResNet-50 and ViT-B/16 satisfy it. The other four improve from P2 to P3 by 0.35–1.04 points. Those reversals are not deleted or relabeled: the corresponding P2−P3 differences are −0.67, −0.72, −1.04, −0.35, +0.52, and +0.02 points in table order, and none has an adjusted confirmatory claim.

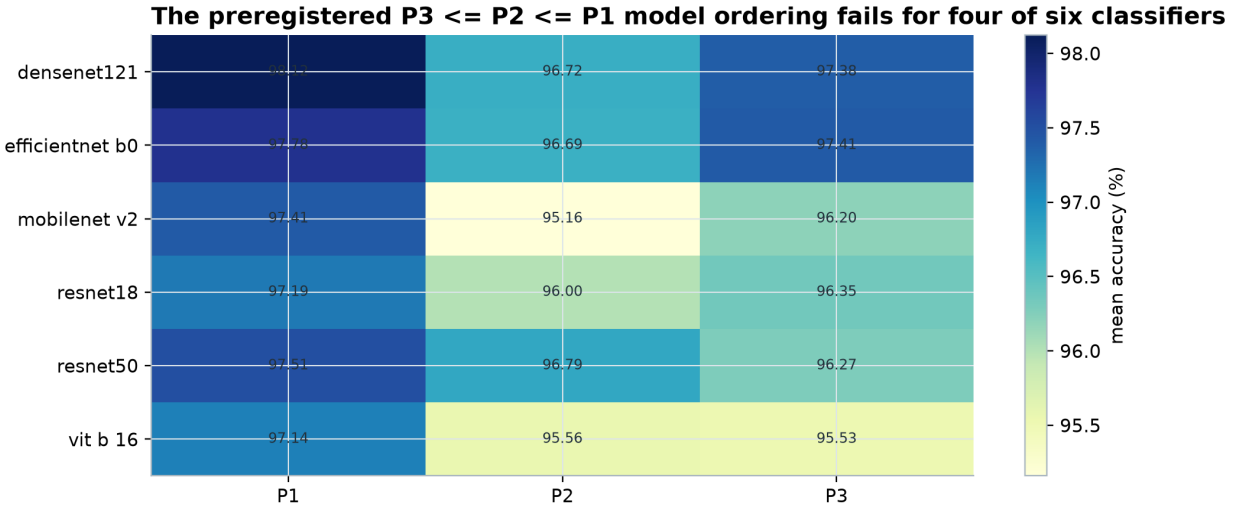


Figure 10.4: The registered ordering check. Cell values are the protocol means; the title records the gate outcome. Four of six architectures fail P3≤P2≤P1, so the acceptance contract correctly records one failed gate.

Architecture ranks are also unstable across P1 and P2: Kendall's $\tau = 0.467$ with bootstrap interval [0.071, 0.733] and $p = 0.2722$; four of the 15 architecture pairs reverse. This is a descriptive warning that a benchmark split can alter conclusions about model selection even when absolute score differences look modest.

10.6 What the classification rescore does and does not show

The same-sign result across six architectures is scientifically suggestive and consistent with the structural mechanism. The five-seed paired variability is nevertheless too large for any architecture to pass the six-test BH family. The correct report is therefore: *crop-disjoint scores were higher on average for every tested classifier, by 0.72–2.25 percentage points, but the preregistered architecture-wise confirmatory family did not establish individual effects after multiplicity correction.*

No classifier confusion matrix appears because the frozen result schema retains aggregate test metrics and epoch histories, but not per-sample predictions or model checkpoints. Generating such a matrix now would require a new experiment after the declared research stop. Omitting it is more faithful than reconstructing or simulating unobserved cells.

10.7 Training behavior and compute

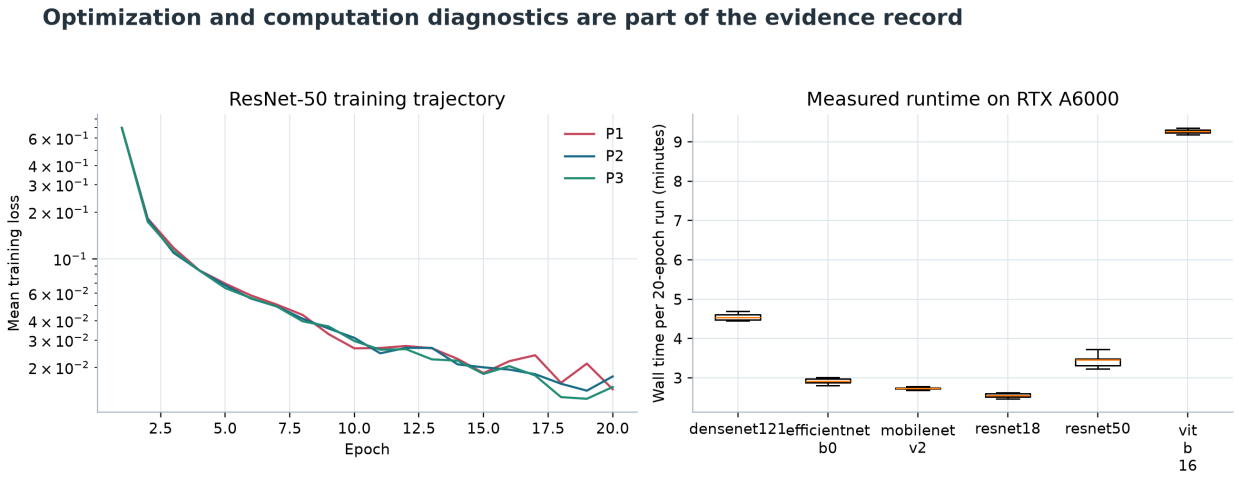


Figure 10.5: Classifier wall-clock summaries from the retained run JSON files. These are same-system measurements and should not be compared to timings reported on unrelated hardware.

The 120 retained classification runs consumed 8.48 summed run-hours. Because up to three jobs ran concurrently, this is computational work, not elapsed calendar duration. Shorter wall time does not imply a more accurate or better-calibrated model; it is included so the execution cost is auditable.

11

Cx22 segmentation rescore

11.1 Why add a second task and dataset?

Classification alone could leave two alternative explanations: perhaps the phenomenon is specific to SIPaKMeD, or perhaps it is specific to assigning a single label to a crop. Cx22 offers a different hierarchy and two pixel-level tasks. It therefore tests whether the split logic transfers to semantic and instance segmentation without pretending that its official split is the same estimand as the pooled stress test.

The pooled analysis treats all 1,320 released generated regions as the observation pool and the 514 CCEDD sources as parents. The protocolOFFICIAL arm uses only Cx22-Multi-Train (400 observations) and Cx22-Multi-Test (100 observations). These official train and test source-parent sets are disjoint. Pair is not added to the official test comparison: 55 of its source parents overlap Multi-Test and 233 overlap Multi-Train, with Jaccard similarities 0.126 and 0.475 respectively. Pooling and re-splitting is thus a downstream-reuse stress test, not an accusation that the authors' official Multi split is flawed.

11.2 Targets and models

The semantic target has two independent binary channels: cytoplasm and nucleus. Five encoder–decoder families are re-run: U-Net [23], U-Net++ [31], DeepLabV3+ [3], FPN [14], and MAnet [5]. ResNet-34 is the encoder except for DeepLabV3+, which uses ResNet-50. All encoders start from ImageNet weights. The output activation is not embedded in the network; logits are passed through sigmoid in the loss and evaluation code.

For image i, channel c, prediction probability p_{icu}, and binary target y_{icu} at pixel u, the training objective is

$$\mathcal{L}_{\text{semantic}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{softDice}}, \tag{11.1}$$

$$\mathcal{L}_{\text{softDice}} = 1 - \frac{1}{2N} \sum_{i,c} \frac{2 \sum_u p_{icu} y_{icu} + 1}{\sum_u p_{icu} + \sum_u y_{icu} + 1}. \tag{11.2}$$

The instance task uses Mask R-CNN with a ResNet-50 FPN backbone and its torchvision V2 implementation [8]. The classification and mask heads are replaced for one foreground category plus background. Ground-truth cytoplasm instances are reconstructed from the released MATLAB label structure before resizing. Empty instances after resizing are rejected rather than silently retained.

11.3 Deterministic input construction

All images and masks are resized to 256×256; RGB images use antialiased interpolation, while masks use nearest-neighbor interpolation. Training augmentation is determined by a SHA-256 digest of seed, epoch, and crop ID. Digest bits choose horizontal flip, vertical flip, and one of four right-angle rotations. This produces per-epoch augmentation without relying on worker scheduling order. Images are ImageNet-normalized; masks remain binary.

The original JPEGs and generated masks were packed into memory-mappable arrays, each named in a provenance manifest with SHA-256 hashes and a cache version. This was not a scientific transformation. It removed repeated JPEG/MATLAB/NFS decoding from the inner loop while making the exact input bytes auditable.

Table 11.1: Cx22 segmentation execution settings.

Item	Semantic	Instance
Image size	256×256	256×256
Epochs	30	30
Batch size	16	2 (test batch 1)
Optimizer	AdamW	AdamW
Learning rate	10^{-4} fixed	10^{-4} fixed
Weight decay	10^{-4}	10^{-4}
Precision	FP16 except MAnet FP32	FP16
Endpoint schedule	every epoch; final retained	final epoch only
Checkpointing	every epoch	every 5 epochs
Seeds	17, 29, 43, 71, 101	17, 29, 43, 71, 101
Hardware	NVIDIA RTX A6000; at most three concurrent GPU jobs	

11.4 Semantic metrics

Probabilities are thresholded at 0.5. For predicted set A and target set B,

$$\mathrm{Dice}(A,B) = \frac{2|A \cap B|}{|A| + |B|}, \qquad \mathrm{IoU}(A,B) = \frac{|A \cap B|}{|A \cup B|}. \tag{11.3}$$

The primary semantic value first averages cytoplasm and nucleus Dice within each image, then averages over test images. This prevents a large cytoplasm region from numerically drowning the smaller nucleus. Channel-specific Dice and IoU are retained as diagnostic endpoints.

11.5 Instance matching metric

Predictions with object score below 0.5 are discarded. Let D_{jk} be Dice between predicted instance j and ground-truth instance k. The Hungarian assignment [12] maximizes

$$\max_{\pi} \sum_{(j,k)\in\pi} D_{jk} \tag{11.4}$$

subject to one-to-one matching. Every unmatched ground-truth instance receives Dice and IoU zero. The primary image value is the mean over that image's ground-truth instances; images are then averaged. Recall@0.70 is the fraction of ground-truth instances whose matched Dice is at least 0.70. This endpoint penalizes missed cells and does not allow an image with many easy instances to dominate the result.

Listing 11.1: Exact logic of the matched-instance endpoint.

```
keep predictions with confidence >= 0.50
threshold predicted and target masks at 0.50
compute all predicted-by-target Dice values on the GPU
solve maximum-weight one-to-one assignment
write matched Dice into a zero vector of length n_targets
average the vector within the image
average image means over the test set
```

11.6 Semantic results

Table 11.2: Cx22 semantic macro-Dice (%), mean ± sample SD over five seeds. The official column uses a different 400/100 sample and is descriptive, not a paired protocol contrast.

Model	P1	P1′	P2	P3	OFFICIAL
DeepLabV3+	88.90±0.45	88.84±0.45	88.64±0.34	88.10±0.38	82.96±1.54
FPN	88.90±0.43	88.97±0.55	88.89±0.48	88.33±0.30	84.43±0.33
MAnet	90.03±0.60	90.07±0.63	89.97±0.45	89.63±0.21	85.50±1.41
U-Net	90.29±0.48	90.32±0.52	89.99±0.52	89.77±0.16	85.64±0.58
U-Net++	90.56±0.51	90.58±0.54	90.38±0.35	90.05±0.21	87.30±1.13

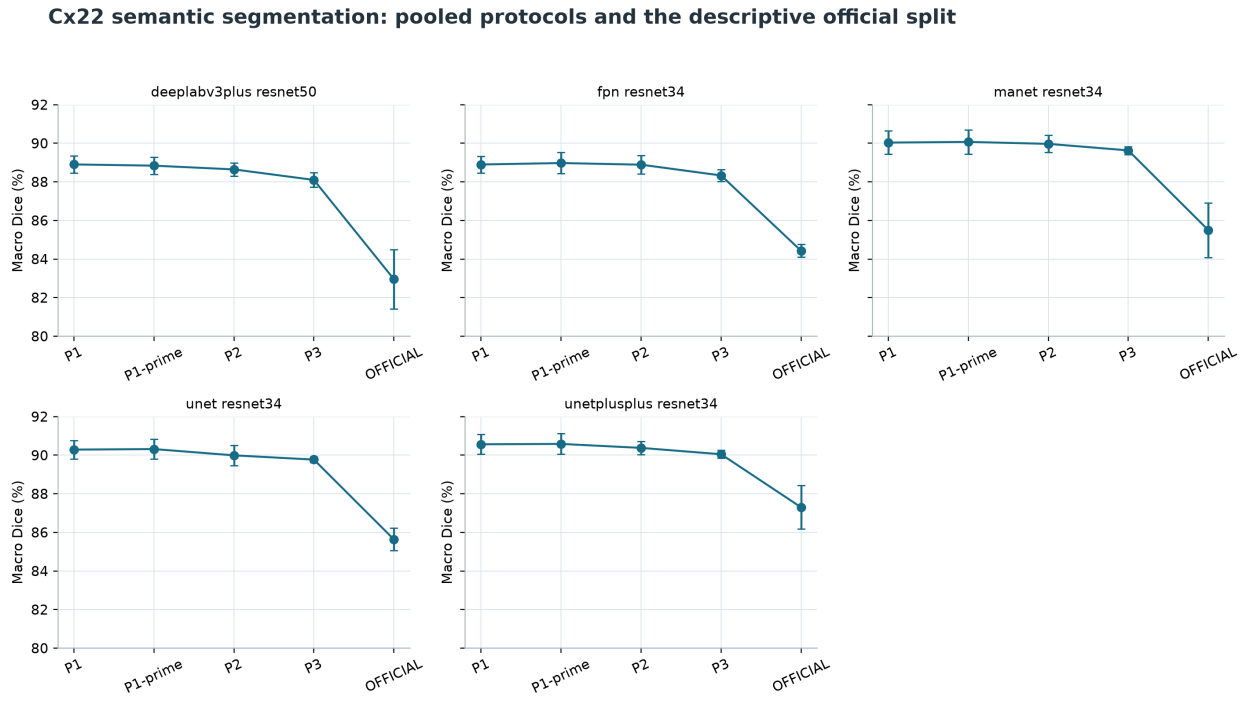


Figure 11.1: Cx22 semantic macro-Dice by architecture and protocol. The official arm is visually separated because it is not a same-observation contrast.

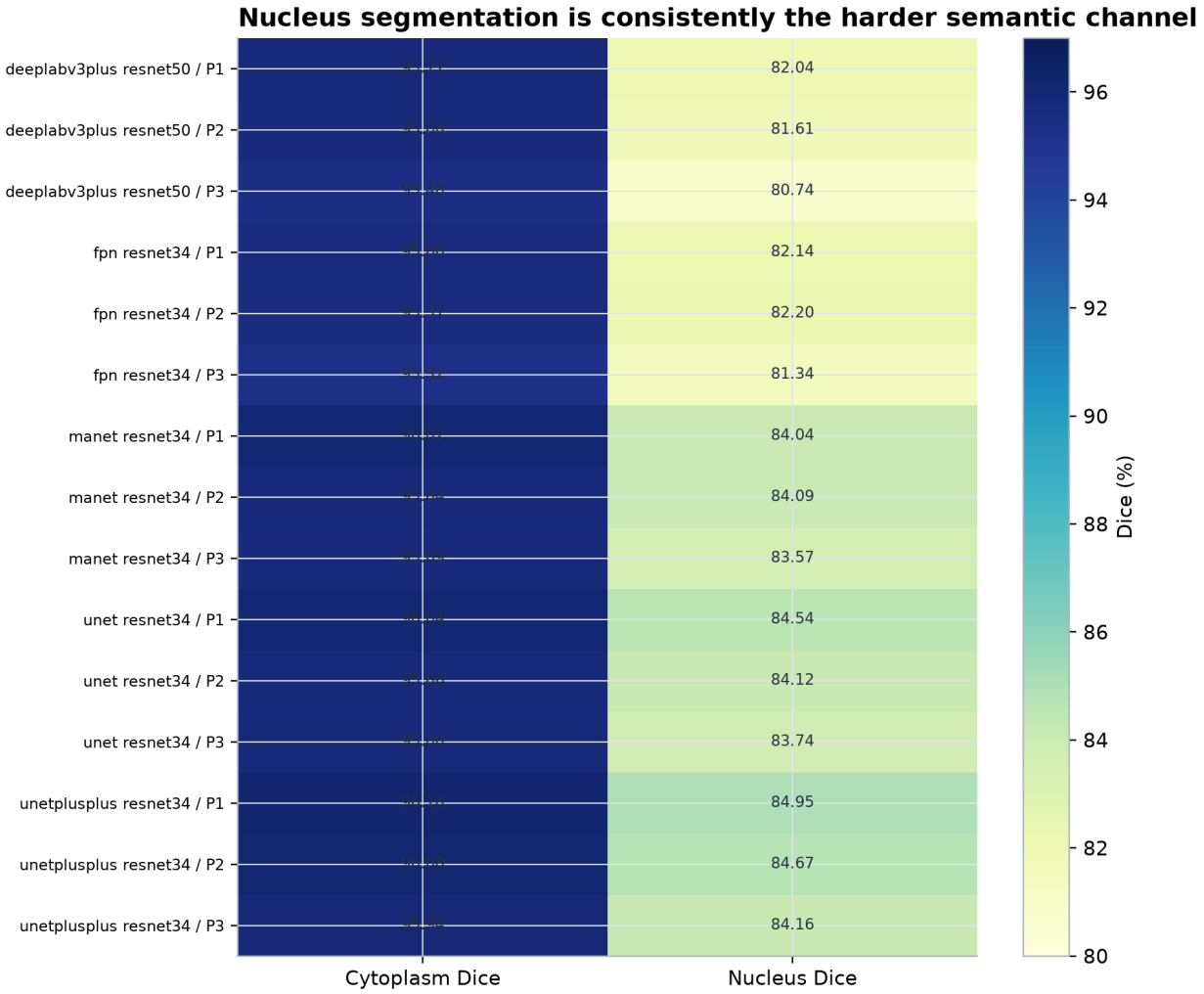


Figure 11.2: Channel-level Dice makes the central difficulty visible: nuclei are consistently harder than cytoplasm. This is the segmentation analogue of a confusion-matrix diagnostic and uses retained aggregate channel measurements, not invented pixel predictions.

U-Net++ is best in every pooled protocol and on the official Multi arm. The channel breakdown explains the lower official macro score: across models, cytoplasm Dice is roughly 94–95% while nucleus Dice is roughly 72–79%. Nuclei are smaller, so a fixed boundary displacement consumes a larger fraction of their area; faint staining and overlap also make the boundary less stable. This is a checkable data property visible in the released masks rather than a generic explanation added after the fact.

11.7 Instance results

Table 11.3: Cx22 matched-instance Dice (%), mean ± sample SD over five seeds.

Model	P1	P1′	P2	P3	OFFICIAL
Mask R-CNN	92.10±0.52	92.05±0.52	91.59±0.42	91.11±1.09	92.29±0.32
Mask R-CNN V2	91.99±0.57	91.95±0.58	91.36±0.82	90.60±0.52	92.18±0.71

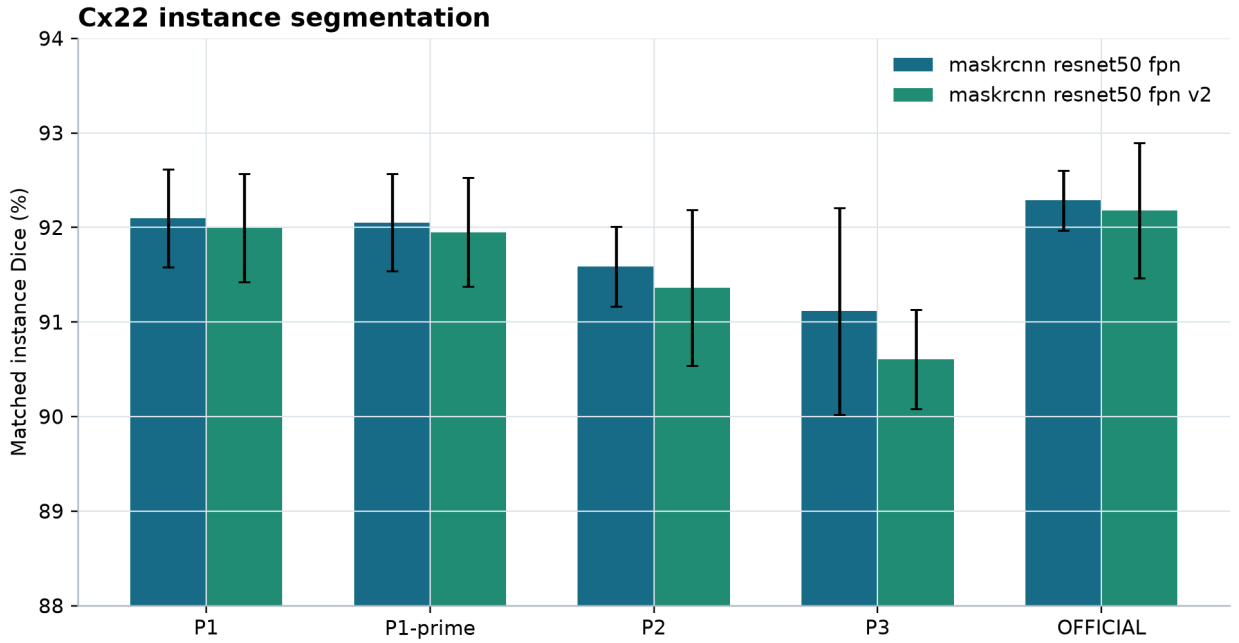


Figure 11.3: Cx22 matched-instance Dice. Both Mask R-CNN variants follow the complete registered P3≤P2≤P1 ordering.

Unlike SIPaKMeD classification, all seven Cx22 architecture/task rows satisfy the registered ordering. The differences are still small: P1−P2 is 0.01–0.30 percentage points for semantic models and 0.51–0.63 points for instance models.

11.8 Inference and rank stability

Table 11.4: Primary Cx22 P1−P2 contrasts. Values are mean ± SD percentage points; q is within the task-specific preregistered family. Effect is Cohen's d_z except DeepLabV3+, whose non-normal seed differences trigger rank-biserial effect size.

Task	Model	Mean±SD	Raw p	BH q	Effect
Semantic	DeepLabV3+	0.258±0.418	0.3125	0.8458	0.600†
Semantic	FPN	0.010±0.468	0.9625	0.9625	0.022
Semantic	MAnet	0.065±0.450	0.7612	0.8458	0.146
Semantic	U-Net	0.297±0.767	0.4353	0.8458	0.387
Semantic	U-Net++	0.188±0.406	0.3582	0.8458	0.464
Instance	Mask R-CNN	0.509±0.581	0.1220	0.2440	0.875
Instance	Mask R-CNN V2	0.630±1.204	0.3068	0.3116	0.523

†rank-biserial correlation; all other effects are Cohen's d_z.

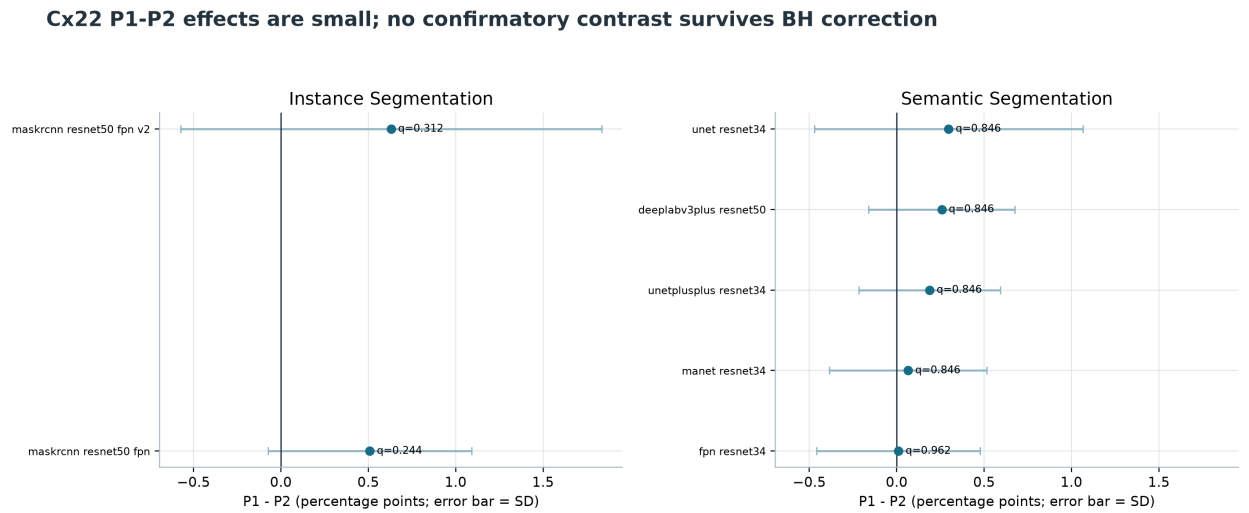


Figure 11.4: Cx22 paired contrast overview. Positive values favor the crop-level protocol. The plot separates point-estimate consistency from multiplicity-adjusted evidence.

No primary Cx22 test survives BH correction. The size-matched $\mathsf{P1}'-\mathsf{P2}$ Mask R-CNN contrast has raw $p = 0.0377$, $d_z = 1.368$, and $q = 0.1508$; it is not promoted to a significant claim. In the explicitly secondary $\mathsf{P2}-\mathsf{P3}$ family, FPN has raw $p = 0.0226$ and $d_z = 1.613$; it remains secondary and is not used to rescue the main conclusion.

Semantic model ranking is comparatively stable: Kendall's $\tau = 0.8$ with bootstrap interval [0.6, 1.0], $p = 0.0833$, and one of ten model pairs reordered. With only two instance models, $\tau = 1.0$ but its bootstrap interval spans $[-1,1]$, which is appropriately uninformative.

11.9 Training failures, recovery, and runtime

The first MAnet P1, seed-17 attempts produced non-finite losses at epochs 23 and 24 under FP16, despite dynamic loss scaling. Retaining an epoch-22 model or changing only that seed would have biased the matrix. The matrix stopped at 100 of 175 cells, the failure was amended in writing, and every MAnet cell was restarted in FP32 from its fixed seed. The corrected runs also set `CUBLAS_WORKSPACE_CONFIG=:4096:8` before any corrected MAnet or instance result existed.

Instance segmentation exposed a different failure: not numerical instability, but pipeline starvation. Per-epoch test inference, CPU pairwise mask matching, and network-filesystem checkpoint writes made GPUs wait. The final endpoint was already preregistered, so evaluation was restricted to the final epoch, exact binary intersections moved to the GPU, checkpoints became five-epoch recovery files, and immutable arrays were staged locally. These changes preserved the endpoint and substantially improved utilization; they did not alter splits, seeds, models, thresholds, or the final metric.

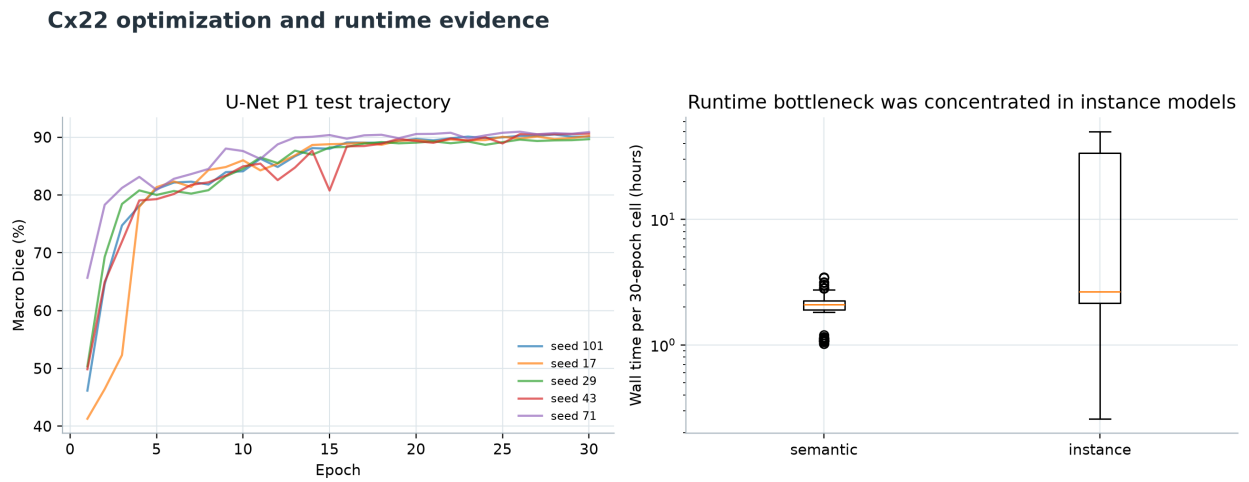


Figure 11.5: Cx22 run-time distribution. Instance segmentation dominates the computational cost and has a long tail; the plot makes the operational bottleneck explicit.

The 125 semantic cells consumed 253.33 summed run-hours, with median cell time 2.09 hours. The 50 instance cells consumed 646.76 summed run-hours, with median 2.64 hours and range 0.258–49.79 hours. The complete Cx22 matrix consumed 900.08 summed run-hours. Up to three jobs were concurrent, so the sum is not wall-clock project duration.

As in classification, final model weights and per-pixel predictions were not retained after the completed JSON was written. Figure 4.3 therefore visualizes real released ground truth, not a simulated qualitative prediction. A post-hoc prediction gallery would require fresh inference and is deliberately outside the frozen record.

12

Statistical analysis in full

12.1 The experimental unit is the seed-paired run

Every reported neural summary uses the five fixed seeds 17, 29, 43, 71, and 101 and gives the arithmetic mean with sample standard deviation ($n - 1$ denominator). The protocol contrast is paired by seed. This pairing removes seed-to-seed variation that affects both arms and prohibits a favorable run in one protocol from being compared with an unrelated run in another.

For values m_s over $S = 5$ seeds,

$$\bar{m} = \frac{1}{S}\sum_{s=1}^{S} m_s, \qquad s_m = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}(m_s - \bar{m})^2}. \tag{12.1}$$

A summary is suppressed if fewer than five seeds are present. This is enforced in code rather than remembered during table writing.

12.2 Test-selection rule

The preregistered rule first applies Shapiro–Wilk [25] to the five paired differences. If $p \geq 0.05$, a two-sided paired t-test is emitted; otherwise a two-sided Wilcoxon signed-rank test [28] is emitted. The effect is Cohen's d_z for a t-test and rank-biserial correlation for Wilcoxon. All-zero differences are descriptive; a constant nonzero difference has an infinite standardized effect by definition and is flagged explicitly.

With only five observations, normality screening has little power and neither test can create information that is not present. The rule is retained because it was specified before results, but conclusions emphasize effect direction, variability, and corrected evidence rather than treating the test label as a guarantee.

12.3 Bootstrap effect intervals

Ten thousand deterministic bootstrap replicates resample the vector of paired differences with replacement. Each replicate recomputes the relevant effect-size estimator. The interval is the 2.5th and 97.5th percentiles of finite estimates. A SHA-256-derived bootstrap seed from the dataset/-model/protocol key makes intervals repeatable without coupling them to the training seed.

These are intervals for the standardized effect, not raw mean-score intervals. Because $n = 5$, they can be wide or asymmetric and can contain values much larger than the point estimate. That apparent awkwardness is a correct display of limited information.

12.4 Multiplicity families

Benjamini–Hochberg [2] is applied once to the exact supplied family:

- SIPaKMeD classification: all six P1−P2 and six P1′−P2 tests in the registered joint family;
- Cx22 semantic: five P1−P2 and five P1′−P2 tests;
- Cx22 instance: two P1−P2 and two P1′−P2 tests.

Secondary P2−P3 tests and ranking stability are clearly labeled secondary/unadjusted and cannot substitute for a failed primary family. For ordered raw values $p_{(1)} \le \cdots \le p_{(M)}$, BH computes monotone adjusted values based on

$$q_{(i)} = \min_{j \ge i} \left(1, \frac{M}{j} p_{(j)}\right). \tag{12.2}$$

12.5 Structural sampling error

The expected parent-straddle probability for a parent of size k and 80/20 crop assignment is exact: $1 - 0.8^k - 0.2^k$. Dataset-level analytic expectations average this quantity over recovered parents. Monte Carlo means and SDs over the same five fixed seeds check the split implementation, while 10,000 additional simulations form the broad analytic envelope recorded in the structural table. Herlev is a stronger software test than a confidence interval: because every parent has $k = 1$, both the analytic and observed straddle fractions must be exactly zero.

12.6 Why a non-significant result is not “no effect”

A BH-adjusted $q > 0.05$ says that the current five-seed family does not meet its declared evidential threshold. It does not prove a zero effect. Conversely, six same-sign classifier means do not license six independent significance claims. The report keeps both facts visible: the direction is consistent, and the adjusted confirmatory evidence is insufficient.

13

Execution lifecycle, failures, and demotivations

13.1 Run inventory

Table 13.1: Frozen neural-run inventory and summed run time.

Experiment	Cells	Retained complete	Summed hours
SIPaKMeD classification	120	120	8.48
SIPaKMeD parent probe	30	30	5.75
Cx22 semantic segmentation	125	125	253.33
Cx22 instance segmentation	50	50	646.76
Total	325	325	914.31

The total is the sum of recorded per-run elapsed seconds. It is intentionally not converted into "GPU-days" without qualification: host I/O stalls and overlapping runs mean elapsed seconds include more than active kernel time, while three concurrent GPUs mean the sum exceeds calendar duration.

13.2 The first classification matrix was invalid

An early classifier attempt used scikit-learn 1.7.2. The resulting P2 split counts disagreed across cells, violating the requirement that every architecture see the same seed-specific observations. Twelve JSON files from that attempt were excluded in full. The environment was pinned to scikit-learn 1.9.0, the expected split hashes and counts were made acceptance checks, and the entire 120-cell matrix restarted. Three additional one-epoch files were smoke tests and were never mixed into the result family.

This failure changed the engineering standard: a result is not accepted merely because training finished. Dataset name, architecture, protocol, seed, train/test counts, library versions, provenance hash, split hash, Git SHA, GPU field, metric finiteness, and history length must all match the matrix contract.

13.3 MAnet numerical failure

MAnet attention matrix multiplications overflowed under FP16 late in training. Two attempts failed non-finitely at epochs 23 and 24 for the first MAnet cell. The tempting choices—use the last finite epoch, lower the learning rate only for that cell, or replace seed 17—would each change the estimand after seeing its behavior. The accepted response was broader: freeze an amendment, run every MAnet cell in FP32, restart from the same seeds, and keep the failures in the record.

13.4 The host bottleneck

The remote machine had three available RTX A6000 GPUs but a separate storage/host bottleneck. Heavy concurrent decoding and checkpoint writes could make all GPUs appear slow and could destabilize the SSH session. Launchers therefore queried memory usage and utilization, admitted no more than three idle devices, used atomic result writes, and kept resumable checkpoints. The instance-segmentation redesign reduced expensive network I/O and moved exact matching work to the GPU.

What did not work

The project did not produce a universal multiplicity-corrected classifier inflation result, did not validate the preregistered classification ordering, did not retain prediction-level classifier outputs for a confusion matrix, and has not yet demonstrated a raw-download-to-Table-1 run on a machine that has never seen the project. Each limitation is part of the result, not a cosmetic omission.

13.5 Research stop and evidence freeze

After the publication verdict, exploration stopped. No architecture was added to chase significance; no seed was replaced; no endpoint was changed; no new confusion matrix was generated. The long-form report reconstructs plots only from frozen aggregates and epoch histories. Images used to explain Otsu/Telea and Cx22 label decoding are implementation demonstrations from locally available released data and are labeled as such.

14

Verification against the acceptance contract

14.1 Gate-by-gate status

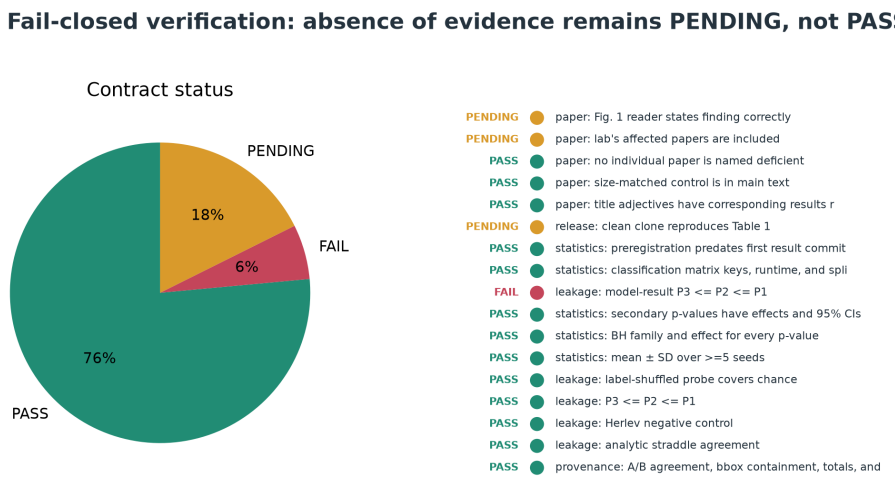


Figure 14.1: Machine-readable acceptance status: 13 gates pass, one fails, and three remain pending. A pending release gate is not painted green simply because the numerical experiments are complete.

Table 14.1: Acceptance contract with evidence.

Gate	Status	Evidence
SIPaKMeD branch agreement $\geq 99\%$	**PASS**	4,049/4,049 mappings agree (100%).
Every bbox lies in parent bounds	**PASS**	4,049/4,049 pass geometry checks.
Crop and per-class parent totals match	**PASS**	4,049 crops; parent vector 126/108/238/271/223.
Observed straddle matches analytic prediction	**PASS**	All audited multi-crop datasets fall inside the recorded simulation envelope.
Herlev has no structural effect	**PASS**	All five seeds return exactly zero in P1; P2/P3 also zero.

Gate	Status	Evidence
P3≤P2≤P1, structural	**PASS**	Grouped protocols eliminate measured parent sharing.
Shuffled probe confirms chance	**PASS**	95% t interval $[-0.1385, 0.4687]$% contains 0.2358%.
Every neural summary has ≥ 5 seeds	**PASS**	120 classifier, 30 probe, and 175 Cx22 files pass matrix completeness.
Plan predates first result commit	**PASS**	22 Aug 2026 04:54:10 IST versus 13:22:42 IST.
BH applied over stated families	**PASS**	Stored family labels and q values are reproduced from the exact supplied rows.
Effect accompanies every inferential p	**PASS**	d_z, rank-biserial, or Kendall τ is present.
Classifier P3≤P2≤P1	**FAIL**	Only ResNet-50 and ViT-B/16 pass; four of six fail.
Cx22 model ordering	**PASS**	All five semantic and two instance rows pass.
Official Cx22 parent disjointness	**PASS**	Multi-Train and Multi-Test have no shared recovered source parent.
Virgin-host clean-clone reproduction	**PENDING**	Script exists, but execution evidence from a never-used host is absent.
Lab's own affected papers in audit	**PENDING**	No verified crosswalk in the frozen repository.
Figure-1-only independent reader	**PENDING**	No documented colleague response in the frozen repository.

14.2 Why the classification ordering gate fails

The structural ordering and model-score ordering are different propositions. P3 has stricter grouping than P2, but it also creates a different balanced partition. Removing a small duplicate-connected component can alter class composition, parent difficulty, and which observations land in test. A finite test set can therefore score higher under the stricter split even though it contains less structural sharing. The failure warns against using monotonic score ordering as a definition of correctness; leakage absence should be checked directly from IDs and components.

14.3 Release-readiness boundary

The repository contains a one-script verification path, but the strongest release criterion requires a truly clean machine that has never held the repository or data. Running the script on the development host, even in a new directory, would not fully establish that environmental condition. This report therefore leaves the release gate pending. The same standard applies to the human communication gate and lab-paper crosswalk: absence of evidence is not converted into a pass.

15

Interpretation, limitations, and practical guidance

15.1 What the evidence supports

The source hierarchy is real and recoverable. Random crop splitting predictably separates siblings. In SIPaKMeD it can also put overlapping source pixels into training and test. Parent identity is learnable even after large parts of the apparent cell signal are removed. Parent- and component-disjoint protocols eliminate the directly measured sharing. These conclusions follow from deterministic identities, geometry, and controls, not from a single favorable neural score.

The downstream effect is subtler. Every tested $\mathsf{P1}-\mathsf{P2}$ mean is positive, including six classifiers and seven segmentation models, but the Cx22 semantic differences are often tiny and none of the task-specific confirmatory families survives BH correction. The evidence supports changing split construction because independence is a design requirement, not because a guaranteed score penalty has been estimated.

15.2 Two hard visual structures in Cx22

The most persistent semantic error source is the nucleus channel. Nuclei occupy fewer pixels, so boundary uncertainty has a larger Dice penalty; pale nuclei and overlaps further reduce contrast. The second hard structure is touching cytoplasm instances. A union mask can remain visually plausible while an instance model merges two touching cells; maximum matching then assigns a good overlap to one target and zero or low overlap to the other. These explanations correspond to the actual label geometry in Figure 4.3 and the channel results in Figure 11.2.

15.3 Real limitations

1. **Dataset hierarchy stops at the released identifier.** SIPaKMeD fields and Cx22/CRIC sources are not guaranteed to be patient IDs. Parent-disjoint is therefore necessary but may not be sufficient for clinical independence.

2. **Five seeds leave broad uncertainty.** The complete matrix is large in GPU hours but small in inferential replication. Shapiro screening and bootstrap effects are fragile at $n = 5$.

3. **No clinical validation.** The study audits public research benchmarks; it does not estimate

screening sensitivity, specificity, prevalence calibration, reader impact, or deployment safety.

4. **Hardware and host constraint.** Execution occurred on one login node with at most three A6000 GPUs and an unstable I/O bottleneck. Wall times are not portable performance claims.
5. **Probe transformations are not surgical interventions.** Grayscale, normalization, inpainting, and high-pass filtering each alter multiple cues. They localize signal classes but do not identify a unique causal artifact.
6. **Near-duplicate grouping is conservative and calibrated synthetically.** It found no exact cross-parent duplicates and validated candidate detection on 3 positive/100 negative synthetic examples, but it is not a proof that every biological near-duplicate was linked.
7. **Prediction-level artifacts were not retained.** Post-hoc classifier confusion matrices and qualitative prediction panels cannot be reconstructed without new inference.
8. **Reproduction remains pending on a virgin host.** Dependency or URL drift may still be exposed by that final test.

15.4 Recommended evaluation rule

For any crop-derived cytology dataset, publish at least four fields for every released observation: a stable observation ID, an acquisition/source parent ID, a patient/specimen ID when ethically and legally possible, and a duplicate-component ID if cross-parent duplication was audited. Construct folds at the highest available independent level and publish the exact fold manifest. Report sample counts at both observation and parent levels. Never compare methods under nominally identical dataset names unless the split identifiers also match.

15.5 What to say in a paper

The defensible headline is: “Random crop-level evaluation in public cervical cytology can place sibling and even pixel-overlapping crops across train and test; source-disjoint manifests remove this non-independence.” A stronger sentence—“all published scores are inflated”—is not supported. No individual prior paper needs to be named as deficient; the target is a reusable benchmark practice.

16

Reproduction and release engineering

16.1 From owner downloads to structural Table 1

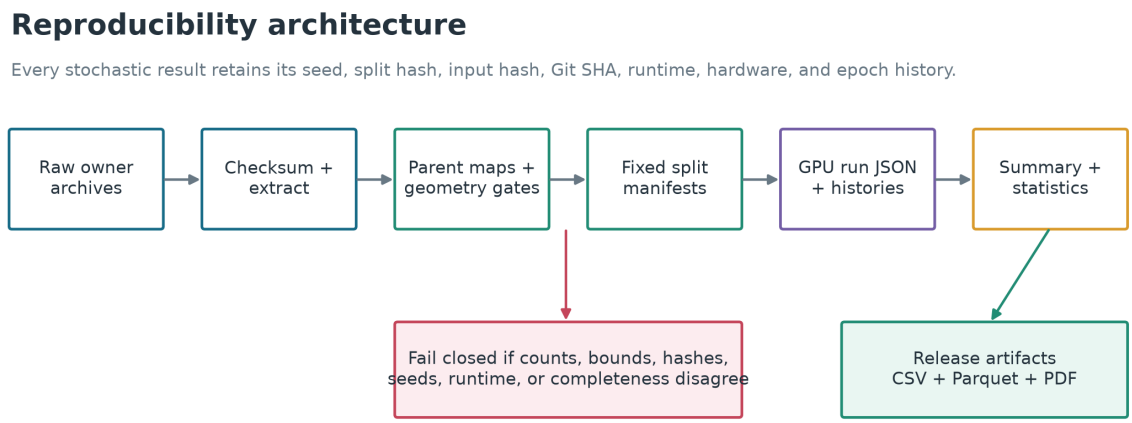


Figure 16.1: The intended clean-clone path. Each boundary writes an auditable artifact and performs validation before the next stage begins.

The GPU-free structural result has one public entry point:

Listing 16.1: Raw-download-to-Table-1 entry point.

```
./scripts/reproduce_table1.sh
```

The script performs a locked dependency sync; downloads all declared owner files with at most three connections; verifies exact sizes and SHA-256 values; extracts archives; recovers SIPaKMeD, Cx22, CRIC, and Herlev provenance; runs the SIPaKMeD duplicate and pixel-overlap audits; creates fixed splits; and emits the dataset-level and seed-level structural tables.

Raw images are intentionally excluded from version control. The repository distributes URLs, checksums, IDs, coordinates, derived maps, split assignments, and results, subject to each source's terms. In particular, Cx22/CCEDD assets are non-commercial research resources and are not repackaged in this report.

16.2 Clean-clone attestation

The wrapper requires a repository URL and a destination that does not yet exist:

Listing 16.2: Clean-clone verification wrapper.

```
./scripts/verify_clean_clone.sh \
  REPOSITORY_URL NEW_DESTINATION [GIT_REF]
```

It uses `gitclone--no-local`, optionally checks out an exact detached ref, runs the reproduction script, and writes an attestation. A human/operator must set `CYTOFAIR_VIRGIN_HOST_ATTESTED=YES` only on a host that genuinely has never seen the project. The current repository lacks that final attested evidence, so release remains pending rather than assumed.

16.3 Pinned software and result metadata

The package requires Python 3.11 or newer. The result contract records NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.1, scikit-learn 1.9.0, Pillow 12.3.0, OpenCV 5.0.0, PyTorch 2.9.0+cu128, torchvision 0.24.0+cu128, and CUDA 12.8 for the accepted classifier matrix. The lock file, not these prose values, is the install authority. The explicit versions remain useful diagnostics when a future lock update changes a hash.

16.4 Atomicity and resumability

Long GPU runs write checkpoints locally and copy them to a temporary destination name before atomic rename. A checkpoint is accepted only when its protocol, architecture, seed, and manifest hash match the requested cell. The launcher skips a result only after validating its completed JSON. A second launch can therefore resume interrupted work without accepting a mismatched or half-written file.

16.5 Human-readable and machine-readable layers

The project separates three layers:

- **canonical data**: Parquet provenance maps and CSV split assignments;
- **machine evidence**: JSON run records, CSV summaries, hashes, and acceptance states;
- **human interpretation**: Markdown reports, figures, the conference paper, and this lifecycle monograph.

This prevents a narrative edit from changing the measured evidence and makes it possible to regenerate prose tables from structured files.

17

Reusable outputs and future work

17.1 What exists now

The most durable asset is not a new image collection. It is a stable interface between released observations and independent units: canonical parent maps, duplicate-component identifiers, fixed split CSVs, adjacent hashes, a leak-checking API, a run JSON schema, and paired statistics. A downstream model joins on `crop_id`; it never infers a split from row order or regenerates folds with a library default.

17.2 Two publication units

The conference unit is the complete provenance/leakage argument, mechanism probe, fixed split release, and matched SIPaKMeD rescore. A journal extension can add a separately preregistered annotation study, expanded cross-dataset validation, a broader detector ensemble, and power-oriented learning curves. These are partitioned future studies, not unreported parts of the present evidence.

17.3 Projects unblocked

Table 17.1: Immediate technical reuse. These rows describe enabled work, not completed future experiments.

Project	Reused asset	Work no longer rebuilt
F3: Cx22 seed robustness	Parent map, fixed seeds, run schema, mean/SD and paired inference	Source grouping, seed ledger, rescore scaffold
F8: cross-dataset transfer	Canonical maps, split API, runtime metadata, label-crosswalk slot	Leakage-safe source folds and experiment provenance
F2: foundation encoders	P1/P1′/P2/P3 files and architecture-neutral summarizer	Fair split comparison, seed accounting, BH/effect reporting

17.4 Pre-specified V2 in fish imagery

Fish-Pak contains three views of the same specimen; Croatian Fish contains frames sampled at one frame per second from continuous BRUV/DOV video. The independent units become specimen and video episode. The planned V2 will: recover those units, compute analytic and observed frame straddling, construct specimen/episode-disjoint manifests, run the identity probe, and re-score fixed baselines under $\mathsf{P1}/\mathsf{P1}'/\mathsf{P2}/\mathsf{P3}$. No fish-domain result is claimed here.

17.5 Adoption goal

If versioned CytoFair manifests become the default evaluation interface, future papers can cite both the original dataset and the split asset while reporting exactly which parent-disjoint partition they used. This is an intended impact pathway, not a citation prediction or an empirical result.

18

Conclusion

This project asked whether a file-level train/test boundary corresponds to an independent acquisition boundary in public cervical cytology. It recovered and validated the hidden observation-to-parent hierarchy across four datasets, quantified the analytic and observed structural sharing, demonstrated literal source-pixel sharing in SIPaKMeD, and showed with a shuffled-label-controlled 424-way probe that parent identity remains strongly learnable after multiple cue-removal transformations. It then re-ran six classifiers and seven segmentation models over five fixed seeds and four matched protocols, preserving effect sizes, bootstrap intervals, multiplicity families, failures, and runtime metadata.

The strongest result is methodological: parent-disjoint evaluation is necessary whenever multiple released observations descend from one acquisition unit. The downstream neural results point in a consistent direction but do not justify a universal multiplicity-corrected accuracy-inflation claim. Four of six classifier architectures also violate the preregistered monotonic score ordering, a useful negative result showing that stricter grouping and lower finite-sample score are not synonymous.

CytoFair turns that lesson into reusable artifacts: provenance maps, duplicate components, fixed splits, hashes, fail-closed validators, run ledgers, and a raw-download structural pipeline. The next scientific step is a separately preregistered specimen/episode-disjoint fish-imagery V2; the next release step is a virgin-host clean-clone attestation. Until the latter, the numerical study is complete but the strongest release gate remains honestly pending.

A

Complete SIPaKMeD classifier run ledger

The following 120 rows are the entire accepted classifier matrix. Accuracy, macro-F1, and balanced accuracy are final-epoch test values. This appendix makes every seed visible and prevents mean-only reporting from hiding an outlying run.

Table A.1: Complete SIPaKMeD classification run ledger. All values are final-epoch results.

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
densenet121	P1	17	3239	810	98.519	98.521	98.527	4.66	3
densenet121	P1	29	3239	810	98.025	98.023	98.027	4.62	2
densenet121	P1	43	3239	810	97.778	97.776	97.803	4.54	1
densenet121	P1	71	3239	810	98.025	98.031	98.018	4.61	3
densenet121	P1	101	3239	810	98.272	98.284	98.276	4.70	2
densenet121	P1-prime	17	3239	810	98.519	98.521	98.527	4.61	1
densenet121	P1-prime	29	3239	810	98.025	98.023	98.027	4.48	3
densenet121	P1-prime	43	3239	810	97.778	97.776	97.803	4.49	2
densenet121	P1-prime	71	3239	810	98.025	98.031	98.018	4.48	1
densenet121	P1-prime	101	3239	810	98.272	98.284	98.276	4.45	3
densenet121	P2	17	3239	810	95.926	95.972	95.946	4.47	2
densenet121	P2	29	3239	810	97.778	97.766	97.798	4.55	1
densenet121	P2	43	3239	810	96.914	96.927	96.906	4.51	3
densenet121	P2	71	3239	810	96.543	96.561	96.566	4.47	2
densenet121	P2	101	3239	810	96.420	96.446	96.395	4.59	1
densenet121	P3	17	3239	810	97.778	97.781	97.774	4.54	3
densenet121	P3	29	3239	810	97.901	97.895	97.904	4.64	2
densenet121	P3	43	3239	810	97.531	97.538	97.553	4.56	1
densenet121	P3	71	3239	810	96.790	96.824	96.801	4.20	3
densenet121	P3	101	3239	810	96.914	96.945	96.894	4.00	2
efficientnet_b0	P1	17	3239	810	97.654	97.659	97.662	2.84	1
efficientnet_b0	P1	29	3239	810	98.272	98.276	98.273	2.89	3
efficientnet_b0	P1	43	3239	810	97.160	97.183	97.177	2.92	2
efficientnet_b0	P1	71	3239	810	97.531	97.540	97.541	2.98	1
efficientnet_b0	P1	101	3239	810	98.272	98.279	98.288	2.87	3
efficientnet_b0	P1-prime	17	3239	810	97.654	97.659	97.662	2.87	2
efficientnet_b0	P1-prime	29	3239	810	98.272	98.276	98.273	3.00	1
efficientnet_b0	P1-prime	43	3239	810	97.160	97.183	97.177	2.96	3
efficientnet_b0	P1-prime	71	3239	810	97.531	97.540	97.541	2.88	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
efficientnet_b0	P1-prime	101	3239	810	98.272	98.279	98.288	2.86	1
efficientnet_b0	P2	17	3239	810	97.778	97.790	97.799	3.01	3
efficientnet_b0	P2	29	3239	810	95.185	95.259	95.163	2.92	2
efficientnet_b0	P2	43	3239	810	98.642	98.641	98.643	2.92	1
efficientnet_b0	P2	71	3239	810	95.432	95.450	95.407	2.97	3
efficientnet_b0	P2	101	3239	810	96.420	96.413	96.422	2.93	2
efficientnet_b0	P3	17	3239	810	98.148	98.144	98.170	2.89	1
efficientnet_b0	P3	29	3239	810	97.901	97.920	97.910	2.98	3
efficientnet_b0	P3	43	3239	810	96.296	96.316	96.325	2.99	2
efficientnet_b0	P3	71	3239	810	97.407	97.428	97.427	2.94	1
efficientnet_b0	P3	101	3239	810	97.284	97.300	97.285	2.80	3
mobilenet_v2	P1	17	3239	810	97.284	97.287	97.278	2.72	2
mobilenet_v2	P1	29	3239	810	97.407	97.406	97.403	2.76	1
mobilenet_v2	P1	43	3239	810	96.914	96.933	96.944	2.74	3
mobilenet_v2	P1	71	3239	810	97.284	97.296	97.289	2.71	2
mobilenet_v2	P1	101	3239	810	98.148	98.149	98.167	2.72	1
mobilenet_v2	P1-prime	17	3239	810	97.284	97.287	97.278	2.69	3
mobilenet_v2	P1-prime	29	3239	810	97.407	97.406	97.403	2.74	2
mobilenet_v2	P1-prime	43	3239	810	96.914	96.933	96.944	2.72	1
mobilenet_v2	P1-prime	71	3239	810	97.284	97.296	97.289	2.71	3
mobilenet_v2	P1-prime	101	3239	810	98.148	98.149	98.167	2.73	2
mobilenet_v2	P2	17	3239	810	96.420	96.432	96.453	2.75	1
mobilenet_v2	P2	29	3239	810	93.704	93.732	93.714	2.73	3
mobilenet_v2	P2	43	3239	810	96.667	96.679	96.637	2.74	2
mobilenet_v2	P2	71	3239	810	93.457	93.486	93.458	2.78	1
mobilenet_v2	P2	101	3239	810	95.556	95.553	95.534	2.77	3
mobilenet_v2	P3	17	3239	810	97.407	97.408	97.433	2.68	2
mobilenet_v2	P3	29	3239	810	96.173	96.163	96.163	2.74	1
mobilenet_v2	P3	43	3239	810	96.667	96.669	96.696	2.74	3
mobilenet_v2	P3	71	3239	810	95.802	95.823	95.813	2.92	2
mobilenet_v2	P3	101	3239	810	94.938	94.930	94.900	2.79	1
resnet18	P1	17	3239	810	97.901	97.900	97.930	2.57	1
resnet18	P1	29	3239	810	97.037	97.028	97.051	2.54	2
resnet18	P1	43	3239	810	96.914	96.922	96.938	2.48	3

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet18	P1	71	3239	810	96.420	96.455	96.410	2.61	1
resnet18	P1	101	3239	810	97.654	97.661	97.674	2.54	2
resnet18	P1-prime	17	3239	810	97.901	97.900	97.930	2.53	3
resnet18	P1-prime	29	3239	810	97.037	97.028	97.051	2.62	1
resnet18	P1-prime	43	3239	810	96.914	96.922	96.938	2.61	2
resnet18	P1-prime	71	3239	810	96.420	96.455	96.410	2.48	3
resnet18	P1-prime	101	3239	810	97.654	97.661	97.674	2.62	1
resnet18	P2	17	3239	810	96.420	96.444	96.447	2.58	2
resnet18	P2	29	3239	810	96.049	96.051	96.058	2.47	3
resnet18	P2	43	3239	810	96.543	96.535	96.502	2.49	1
resnet18	P2	71	3239	810	96.543	96.563	96.528	2.56	2
resnet18	P2	101	3239	810	94.444	94.436	94.384	2.50	3
resnet18	P3	17	3239	810	98.395	98.395	98.400	2.55	1
resnet18	P3	29	3239	810	95.062	95.069	95.014	2.55	3
resnet18	P3	43	3239	810	96.173	96.165	96.215	2.52	2
resnet18	P3	71	3239	810	96.543	96.578	96.561	2.79	1
resnet18	P3	101	3239	810	95.556	95.576	95.526	2.79	3
resnet50	P1	17	3239	810	97.778	97.781	97.792	3.73	2
resnet50	P1	29	3239	810	97.284	97.304	97.275	3.29	1
resnet50	P1	43	3239	810	96.790	96.801	96.805	3.32	3
resnet50	P1	71	3239	810	97.901	97.903	97.904	3.48	2
resnet50	P1	101	3239	810	97.778	97.776	97.777	3.45	1
resnet50	P1-prime	17	3239	810	97.778	97.781	97.792	3.26	3
resnet50	P1-prime	29	3239	810	97.284	97.304	97.275	3.45	2
resnet50	P1-prime	43	3239	810	96.790	96.801	96.805	3.52	1
resnet50	P1-prime	71	3239	810	97.901	97.903	97.904	3.34	3
resnet50	P1-prime	101	3239	810	97.778	97.776	97.777	3.54	2
resnet50	P2	17	3239	810	97.037	97.067	97.053	3.47	1
resnet50	P2	29	3239	810	96.914	96.931	96.916	3.24	3
resnet50	P2	43	3239	810	97.284	97.298	97.290	3.45	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet50	P2	71	3239	810	96.420	96.422	96.433	3.47	1
resnet50	P2	101	3239	810	96.296	96.303	96.288	3.29	3
resnet50	P3	17	3239	810	96.543	96.519	96.573	3.47	2
resnet50	P3	29	3239	810	97.037	97.071	97.049	3.46	1
resnet50	P3	43	3239	810	95.679	95.701	95.703	3.25	3
resnet50	P3	71	3239	810	95.926	95.945	95.958	3.73	2
resnet50	P3	101	3239	810	96.173	96.168	96.161	3.89	1
vit_b_16	P1	17	3239	810	97.407	97.414	97.430	9.30	3
vit_b_16	P1	29	3239	810	97.037	97.050	97.061	9.25	2
vit_b_16	P1	43	3239	810	95.556	95.541	95.606	9.35	1
vit_b_16	P1	71	3239	810	98.025	98.036	98.023	9.31	3
vit_b_16	P1	101	3239	810	97.654	97.666	97.684	9.27	2
vit_b_16	P1-prime	17	3239	810	97.407	97.414	97.430	9.29	1
vit_b_16	P1-prime	29	3239	810	97.037	97.050	97.061	9.19	3
vit_b_16	P1-prime	43	3239	810	95.556	95.541	95.606	9.24	2
vit_b_16	P1-prime	71	3239	810	98.025	98.036	98.023	9.30	1
vit_b_16	P1-prime	101	3239	810	97.654	97.666	97.684	9.24	3
vit_b_16	P2	17	3239	810	95.556	95.618	95.526	9.27	2
vit_b_16	P2	29	3239	810	95.802	95.821	95.788	9.36	1
vit_b_16	P2	43	3239	810	96.667	96.693	96.688	9.32	3
vit_b_16	P2	71	3239	810	95.432	95.432	95.392	9.30	2
vit_b_16	P2	101	3239	810	94.321	94.421	94.283	9.28	1
vit_b_16	P3	17	3239	810	96.296	96.258	96.328	9.19	3
vit_b_16	P3	29	3239	810	95.185	95.218	95.214	9.23	2
vit_b_16	P3	43	3239	810	92.840	92.915	92.770	9.26	1
vit_b_16	P3	71	3239	810	97.284	97.292	97.301	9.00	3
vit_b_16	P3	101	3239	810	96.049	96.039	96.066	8.81	2

B

Complete parent-probe run ledger

These 30 rows include five preprocessing arms and the label-shuffled negative control. All use exactly 424 parent classes, 848 training crops, and 848 held-out sibling crops.

Table B.1: Complete parent-identity probe ledger.

Mode	Shuffle	Seed	Parents	Train	Test	Chance (percent)	Accuracy (percent)	min	GPU
background_only	no	17	424	848	848	0.2358	18.514	21.00	3
background_only	no	29	424	848	848	0.2358	20.047	20.55	1
background_only	no	43	424	848	848	0.2358	19.811	20.23	2
background_only	no	71	424	848	848	0.2358	17.925	89.59	3
background_only	no	101	424	848	848	0.2358	17.099	89.40	2
grayscale	no	17	424	848	848	0.2358	15.212	0.93	3
grayscale	no	29	424	848	848	0.2358	13.208	0.88	2
grayscale	no	43	424	848	848	0.2358	14.741	1.02	1
grayscale	no	71	424	848	848	0.2358	13.208	0.97	2
grayscale	no	101	424	848	848	0.2358	12.618	1.02	3
highpass	no	17	424	848	848	0.2358	16.863	15.73	1
highpass	no	29	424	848	848	0.2358	16.156	15.87	1
highpass	no	43	424	848	848	0.2358	16.863	15.93	1
highpass	no	71	424	848	848	0.2358	16.038	15.80	1
highpass	no	101	424	848	848	0.2358	15.448	16.04	1
rgb	no	17	424	848	848	0.2358	25.943	0.89	1
rgb	no	29	424	848	848	0.2358	26.769	0.76	2
rgb	no	43	424	848	848	0.2358	25.236	0.76	3
rgb	no	71	424	848	848	0.2358	25.590	0.77	1
rgb	no	101	424	848	848	0.2358	25.118	0.79	2
rgb	yes	17	424	848	848	0.2358	0.590	0.78	3
rgb	yes	29	424	848	848	0.2358	0.000	0.88	1
rgb	yes	43	424	848	848	0.2358	0.118	0.86	3
rgb	yes	71	424	848	848	0.2358	0.000	0.76	2
rgb	yes	101	424	848	848	0.2358	0.118	0.82	1
stain_normalized	no	17	424	848	848	0.2358	17.335	2.49	1
stain_normalized	no	29	424	848	848	0.2358	17.099	2.41	2
stain_normalized	no	43	424	848	848	0.2358	18.396	2.26	3
stain_normalized	no	71	424	848	848	0.2358	16.745	2.42	1
stain_normalized	no	101	424	848	848	0.2358	16.981	2.20	2

C

Complete Cx22 model run ledger

The following 175 rows are all accepted Cx22 cells: five semantic architectures and two instance architectures, each under five protocols and five seeds. "Primary" is semantic macro-Dice or matched-instance Dice; "Nuc./R70" is nucleus Dice for semantic rows and Recall@0.70 for instance rows.

Table C.1: Complete Cx22 run ledger. Primary is macro Dice for semantic models and matched-instance Dice for instance models; the ninth column is nucleus Dice or recall at Dice 0.70.

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn	OFFICIAL	17	400	100	92.117	87.060	96.991	14.35
instance	maskrcnn_resnet50_fpn	OFFICIAL	29	400	100	92.372	87.354	97.068	14.92
instance	maskrcnn_resnet50_fpn	OFFICIAL	43	400	100	91.844	86.861	96.682	15.89
instance	maskrcnn_resnet50_fpn	OFFICIAL	71	400	100	92.669	87.541	97.608	0.49
instance	maskrcnn_resnet50_fpn	OFFICIAL	101	400	100	92.433	87.559	97.145	1.29
instance	maskrcnn_resnet50_fpn	P1	17	1056	264	92.683	87.370	96.540	33.85
instance	maskrcnn_resnet50_fpn	P1	29	1056	264	91.354	86.327	95.009	33.71
instance	maskrcnn_resnet50_fpn	P1	43	1056	264	92.282	87.191	96.854	35.84
instance	maskrcnn_resnet50_fpn	P1	71	1056	264	92.343	87.320	96.550	37.54
instance	maskrcnn_resnet50_fpn	P1	101	1056	264	91.819	86.735	96.471	2.55
instance	maskrcnn_resnet50_fpn	P1-prime	17	1056	264	92.683	87.370	96.540	33.94
instance	maskrcnn_resnet50_fpn	P1-prime	29	1056	264	91.354	86.327	95.009	33.95
instance	maskrcnn_resnet50_fpn	P1-prime	43	1056	264	92.282	87.191	96.854	35.19
instance	maskrcnn_resnet50_fpn	P1-prime	71	1055	264	91.733	86.958	95.483	38.29
instance	maskrcnn_resnet50_fpn	P1-prime	101	1053	264	92.213	87.257	97.248	2.56
instance	maskrcnn_resnet50_fpn	P2	17	1056	264	91.815	86.503	95.983	33.84
instance	maskrcnn_resnet50_fpn	P2	29	1056	264	91.313	86.228	95.159	34.29
instance	maskrcnn_resnet50_fpn	P2	43	1056	264	92.024	86.559	96.750	35.88
instance	maskrcnn_resnet50_fpn	P2	71	1055	265	90.993	85.998	92.239	0.26
instance	maskrcnn_resnet50_fpn	P2	101	1053	267	91.793	86.639	94.671	2.57
instance	maskrcnn_resnet50_fpn	P3	17	891	429	89.855	84.435	94.937	47.44
instance	maskrcnn_resnet50_fpn	P3	29	825	495	91.606	86.357	95.207	49.79
instance	maskrcnn_resnet50_fpn	P3	43	894	426	90.023	84.807	94.415	47.25
instance	maskrcnn_resnet50_fpn	P3	71	773	547	92.181	86.751	96.205	1.16
instance	maskrcnn_resnet50_fpn	P3	101	791	529	91.910	86.399	96.806	1.98
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	17	400	100	91.970	87.274	95.910	1.31
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	29	400	100	92.548	87.588	96.991	1.33
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	43	400	100	91.041	86.290	94.367	1.32
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	71	400	100	92.442	87.459	97.068	1.69
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	101	400	100	92.886	87.943	97.531	1.40
instance	maskrcnn_resnet50_fpn_v2	P1	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	29	1056	264	91.543	86.726	92.276	2.64
instance	maskrcnn_resnet50_fpn_v2	P1	43	1056	264	92.536	87.688	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P1	71	1056	264	92.342	87.566	94.856	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	101	1056	264	92.318	87.546	95.131	2.99
instance	maskrcnn_resnet50_fpn_v2	P1-prime	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	29	1056	264	91.543	86.726	92.276	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	43	1056	264	92.536	87.688	94.870	2.64
instance	maskrcnn_resnet50_fpn_v2	P1-prime	71	1055	264	91.938	87.241	93.915	2.62
instance	maskrcnn_resnet50_fpn_v2	P1-prime	101	1053	264	92.502	87.584	95.272	2.85
instance	maskrcnn_resnet50_fpn_v2	P2	17	1056	264	92.036	87.121	95.303	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	29	1056	264	91.737	86.754	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P2	43	1056	264	92.056	86.980	95.297	2.65

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn_v2	P2	71	1055	265	90.232	85.363	87.492	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	101	1053	267	90.762	85.672	92.038	2.70
instance	maskrcnn_resnet50_fpn_v2	P3	17	891	429	91.242	85.867	94.762	2.26
instance	maskrcnn_resnet50_fpn_v2	P3	29	825	495	89.960	84.956	91.273	2.10
instance	maskrcnn_resnet50_fpn_v2	P3	43	894	426	90.317	85.094	91.622	2.29
instance	maskrcnn_resnet50_fpn_v2	P3	71	773	547	90.474	85.440	90.769	1.98
instance	maskrcnn_resnet50_fpn_v2	P3	101	791	529	91.026	85.770	94.358	2.02
semantic	deeplabv3plus_resnet50	OFFICIAL	17	400	100	82.697	72.771	71.084	1.02
semantic	deeplabv3plus_resnet50	OFFICIAL	29	400	100	83.602	73.897	72.700	1.13
semantic	deeplabv3plus_resnet50	OFFICIAL	43	400	100	84.126	74.580	73.775	1.08
semantic	deeplabv3plus_resnet50	OFFICIAL	71	400	100	83.972	74.385	73.418	1.12
semantic	deeplabv3plus_resnet50	OFFICIAL	101	400	100	80.382	70.180	66.995	1.11
semantic	deeplabv3plus_resnet50	P1	17	1056	264	88.570	80.909	81.412	2.09
semantic	deeplabv3plus_resnet50	P1	29	1056	264	88.464	80.728	81.373	1.91
semantic	deeplabv3plus_resnet50	P1	43	1056	264	89.196	81.695	82.572	2.02
semantic	deeplabv3plus_resnet50	P1	71	1056	264	89.542	82.165	83.162	2.14
semantic	deeplabv3plus_resnet50	P1	101	1056	264	88.738	81.195	81.662	2.32
semantic	deeplabv3plus_resnet50	P1-prime	17	1056	264	88.570	80.909	81.412	2.13
semantic	deeplabv3plus_resnet50	P1-prime	29	1056	264	88.464	80.728	81.373	1.93
semantic	deeplabv3plus_resnet50	P1-prime	43	1056	264	89.196	81.695	82.572	2.03
semantic	deeplabv3plus_resnet50	P1-prime	71	1055	264	89.443	82.040	82.958	2.14
semantic	deeplabv3plus_resnet50	P1-prime	101	1053	264	88.514	80.848	81.162	2.28
semantic	deeplabv3plus_resnet50	P2	17	1056	264	88.304	80.549	81.070	1.90
semantic	deeplabv3plus_resnet50	P2	29	1056	264	88.930	81.407	82.016	2.03
semantic	deeplabv3plus_resnet50	P2	43	1056	264	88.731	80.967	81.799	1.99
semantic	deeplabv3plus_resnet50	P2	71	1055	265	88.988	81.513	82.202	2.22
semantic	deeplabv3plus_resnet50	P2	101	1053	267	88.268	80.470	80.952	2.28
semantic	deeplabv3plus_resnet50	P3	17	891	429	88.332	80.528	81.111	2.13
semantic	deeplabv3plus_resnet50	P3	29	825	495	87.707	79.682	80.088	2.57
semantic	deeplabv3plus_resnet50	P3	43	894	426	88.621	80.940	81.705	2.70
semantic	deeplabv3plus_resnet50	P3	71	773	547	88.041	80.050	80.594	3.46
semantic	deeplabv3plus_resnet50	P3	101	791	529	87.790	79.677	80.185	3.43
semantic	fpn_resnet34	OFFICIAL	17	400	100	84.378	74.779	74.746	1.08
semantic	fpn_resnet34	OFFICIAL	29	400	100	84.801	75.377	75.422	1.12
semantic	fpn_resnet34	OFFICIAL	43	400	100	84.389	74.760	74.723	1.10
semantic	fpn_resnet34	OFFICIAL	71	400	100	84.630	75.136	75.152	1.07
semantic	fpn_resnet34	OFFICIAL	101	400	100	83.934	74.231	73.850	1.09
semantic	fpn_resnet34	P1	17	1056	264	88.606	80.936	81.657	2.30
semantic	fpn_resnet34	P1	29	1056	264	88.376	80.651	81.204	2.17
semantic	fpn_resnet34	P1	43	1056	264	89.242	81.731	82.802	2.29
semantic	fpn_resnet34	P1	71	1056	264	89.415	81.991	83.139	2.29
semantic	fpn_resnet34	P1	101	1056	264	88.854	81.286	81.903	2.31
semantic	fpn_resnet34	P1-prime	17	1056	264	88.606	80.936	81.657	2.23
semantic	fpn_resnet34	P1-prime	29	1056	264	88.376	80.651	81.204	2.19
semantic	fpn_resnet34	P1-prime	43	1056	264	89.242	81.731	82.802	2.30
semantic	fpn_resnet34	P1-prime	71	1055	264	89.760	82.473	83.695	2.30
semantic	fpn_resnet34	P1-prime	101	1053	264	88.880	81.350	81.891	2.26

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	fpn_resnet34	P2	17	1056	264	88.539	80.901	81.452	2.18
semantic	fpn_resnet34	P2	29	1056	264	89.006	81.470	82.385	2.20
semantic	fpn_resnet34	P2	43	1056	264	89.498	82.080	83.321	2.22
semantic	fpn_resnet34	P2	71	1055	265	89.102	81.618	82.649	2.24
semantic	fpn_resnet34	P2	101	1053	267	88.296	80.471	81.211	2.25
semantic	fpn_resnet34	P3	17	891	429	88.029	80.133	80.698	2.84
semantic	fpn_resnet34	P3	29	825	495	88.080	80.127	80.967	3.16
semantic	fpn_resnet34	P3	43	894	426	88.744	81.048	82.174	2.75
semantic	fpn_resnet34	P3	71	773	547	88.510	80.711	81.583	3.46
semantic	fpn_resnet34	P3	101	791	529	88.287	80.367	81.266	3.45
semantic	manet_resnet34	OFFICIAL	17	400	100	83.189	73.517	71.864	1.06
semantic	manet_resnet34	OFFICIAL	29	400	100	86.409	77.571	77.856	1.08
semantic	manet_resnet34	OFFICIAL	43	400	100	86.521	77.608	78.266	1.07
semantic	manet_resnet34	OFFICIAL	71	400	100	86.256	77.306	77.825	1.15
semantic	manet_resnet34	OFFICIAL	101	400	100	85.118	75.878	75.519	1.19
semantic	manet_resnet34	P1	17	1056	264	89.317	81.998	82.837	2.13
semantic	manet_resnet34	P1	29	1056	264	89.510	82.294	83.125	2.19
semantic	manet_resnet34	P1	43	1056	264	90.535	83.620	85.003	2.09
semantic	manet_resnet34	P1	71	1056	264	90.674	83.752	85.088	2.06
semantic	manet_resnet34	P1	101	1056	264	90.128	83.049	84.158	2.14
semantic	manet_resnet34	P1-prime	17	1056	264	89.317	81.998	82.837	2.17
semantic	manet_resnet34	P1-prime	29	1056	264	89.510	82.294	83.125	2.06
semantic	manet_resnet34	P1-prime	43	1056	264	90.535	83.620	85.003	2.05
semantic	manet_resnet34	P1-prime	71	1055	264	90.764	83.957	85.467	2.05
semantic	manet_resnet34	P1-prime	101	1053	264	90.221	83.230	84.212	2.07
semantic	manet_resnet34	P2	17	1056	264	89.337	81.962	83.086	2.14
semantic	manet_resnet34	P2	29	1056	264	90.185	83.160	84.409	2.08
semantic	manet_resnet34	P2	43	1056	264	90.170	83.032	84.460	2.01
semantic	manet_resnet34	P2	71	1055	265	90.458	83.608	84.919	2.09
semantic	manet_resnet34	P2	101	1053	267	89.687	82.410	83.599	2.00
semantic	manet_resnet34	P3	17	891	429	89.762	82.615	83.748	2.75
semantic	manet_resnet34	P3	29	825	495	89.392	82.029	83.305	2.62
semantic	manet_resnet34	P3	43	894	426	89.895	82.761	83.970	2.16
semantic	manet_resnet34	P3	71	773	547	89.469	82.176	83.199	3.03
semantic	manet_resnet34	P3	101	791	529	89.625	82.349	83.609	1.96
semantic	unet_resnet34	OFFICIAL	17	400	100	85.332	76.145	75.734	1.03
semantic	unet_resnet34	OFFICIAL	29	400	100	85.420	76.382	75.839	1.11
semantic	unet_resnet34	OFFICIAL	43	400	100	85.102	75.810	75.351	1.03
semantic	unet_resnet34	OFFICIAL	71	400	100	86.580	77.740	78.197	1.13
semantic	unet_resnet34	OFFICIAL	101	400	100	85.749	76.664	76.744	1.06
semantic	unet_resnet34	P1	17	1056	264	90.196	83.235	84.389	2.09
semantic	unet_resnet34	P1	29	1056	264	89.654	82.414	83.452	2.16
semantic	unet_resnet34	P1	43	1056	264	90.616	83.797	85.081	2.01
semantic	unet_resnet34	P1	71	1056	264	90.880	84.082	85.657	2.09
semantic	unet_resnet34	P1	101	1056	264	90.093	82.967	84.100	2.04
semantic	unet_resnet34	P1-prime	17	1056	264	90.196	83.235	84.389	2.12
semantic	unet_resnet34	P1-prime	29	1056	264	89.654	82.414	83.452	2.20

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	unet_resnet34	P1-prime	43	1056	264	90.616	83.797	85.081	1.98
semantic	unet_resnet34	P1-prime	71	1055	264	91.020	84.317	85.849	2.06
semantic	unet_resnet34	P1-prime	101	1053	264	90.092	83.064	84.113	1.95
semantic	unet_resnet34	P2	17	1056	264	89.429	82.217	83.000	2.10
semantic	unet_resnet34	P2	29	1056	264	90.249	83.272	84.602	2.17
semantic	unet_resnet34	P2	43	1056	264	90.674	83.773	85.275	2.03
semantic	unet_resnet34	P2	71	1055	265	89.519	82.235	83.280	2.16
semantic	unet_resnet34	P2	101	1053	267	90.082	82.951	84.446	2.01
semantic	unet_resnet34	P3	17	891	429	89.911	82.788	83.948	2.23
semantic	unet_resnet34	P3	29	825	495	89.629	82.392	83.516	2.85
semantic	unet_resnet34	P3	43	894	426	89.892	82.748	83.984	2.82
semantic	unet_resnet34	P3	71	773	547	89.840	82.694	83.704	3.12
semantic	unet_resnet34	P3	101	791	529	89.576	82.274	83.557	2.96
semantic	unetplusplus_resnet34	OFFICIAL	17	400	100	85.933	77.036	76.615	1.06
semantic	unetplusplus_resnet34	OFFICIAL	29	400	100	86.864	78.288	78.526	1.15
semantic	unetplusplus_resnet34	OFFICIAL	43	400	100	86.771	78.140	78.396	1.11
semantic	unetplusplus_resnet34	OFFICIAL	71	400	100	88.276	79.991	81.686	1.08
semantic	unetplusplus_resnet34	OFFICIAL	101	400	100	88.645	80.557	82.123	1.08
semantic	unetplusplus_resnet34	P1	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1	29	1056	264	89.890	82.832	83.858	1.88
semantic	unetplusplus_resnet34	P1	43	1056	264	90.909	84.191	85.525	2.05
semantic	unetplusplus_resnet34	P1	71	1056	264	91.190	84.587	86.038	1.89
semantic	unetplusplus_resnet34	P1	101	1056	264	90.564	83.715	84.868	2.26
semantic	unetplusplus_resnet34	P1-prime	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1-prime	29	1056	264	89.890	82.832	83.858	1.85
semantic	unetplusplus_resnet34	P1-prime	43	1056	264	90.909	84.191	85.525	2.03
semantic	unetplusplus_resnet34	P1-prime	71	1055	264	91.266	84.711	86.229	1.85
semantic	unetplusplus_resnet34	P1-prime	101	1053	264	90.576	83.692	84.939	2.27
semantic	unetplusplus_resnet34	P2	17	1056	264	90.077	83.064	84.058	1.87
semantic	unetplusplus_resnet34	P2	29	1056	264	90.324	83.431	84.639	1.91
semantic	unetplusplus_resnet34	P2	43	1056	264	90.205	83.068	84.221	2.03
semantic	unetplusplus_resnet34	P2	71	1055	265	90.977	84.343	85.735	1.81
semantic	unetplusplus_resnet34	P2	101	1053	267	90.300	83.261	84.697	2.02
semantic	unetplusplus_resnet34	P3	17	891	429	90.011	82.987	84.055	2.10
semantic	unetplusplus_resnet34	P3	29	825	495	89.719	82.531	83.578	2.58
semantic	unetplusplus_resnet34	P3	43	894	426	90.126	83.176	84.281	2.19
semantic	unetplusplus_resnet34	P3	71	773	547	90.283	83.365	84.578	2.56
semantic	unetplusplus_resnet34	P3	101	791	529	90.099	83.076	84.302	2.83

D

Owner data files and integrity hashes

This table is generated from `data_sources.json`. A reproduction refuses a file whose byte length or SHA-256 differs, even if its archive name is unchanged.

Table D.1: Declared owner-source files, exact byte sizes, checksums, and use notes.

Dataset	File	Bytes	SHA-256	Terms note
sipakmed	im_Superficial-Intermediate.7z	762,939,737	3b73e406df14d60b1bd9c8ac3b01b594e70bb51ba32fb080141fc6a7f489a599	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Parabasal.7z	548,667,699	13e9c29f016a5b83731e038cfe518671a7815df28836dee706acba6ecd2369a8	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Koilocytotic.7z	1,290,478,065	653578e878e3cc63a8510cd7ec26d918637d9881e913e29a5bb205d0d9cfb70d	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Metaplastic.7z	1,396,480,123	97f1badc5f606f127d2af0d18a008ebd41a2b60f44785cb332a6eaa33004d2af	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Dyskeratotic.7z	1,182,980,549	a8768bc03c9e814063ff946b7ef7548067e16b760168267d2b12e95a66955cce	Public for experimental use; cite Plissiti et al., ICIP 2018
cx22	Cx22-Multi-Test.zip	1,780,461	6d444e6e4bf8734073bc90e6e2bb29b0e36c66f7b783cbad9f969208621624bb	Non-commercial research and education only
cx22	Cx22-Multi-Train.zip	6,361,959	171be9caf6d0709f3fce2b81a07e1974229c5faae355f197efe553ec1d6d1992	Non-commercial research and education only
cx22	Cx22-Pair.zip	3,705,743	130b6a92db9811151a60e27dc70f4180965ecf0f1dbc84146c87d00c41d7ba9f	Non-commercial research and education only
ccedd	CCEDD.zip	1,507,832,267	d85e6a380a15923f0c2b38e13dd31467f75d11b49e8d21740f16ee69082cdcb1	Non-commercial research and education only; redistribution prohibited
cric	classifications.csv	1,224,400	e1f50183998ddb7392d2a0d4c96dce44c00dbe3833df03f266cc36e047445de5	CC BY 4.0; cite Rezende et al. and the Figshare record
herlev	Cervical Cell Image Database.zip	44,688,829	5b5b740abb599710b9d97fe72954540df9ca7255b2b40a52c8b25ac90d0c50da	CC BY 4.0; repackaging of the original Herlev cervical-cell images

E

Released split hashes

Each CSV contains crop ID, parent ID, partition, protocol, and seed. Hashes identify the exact semantic assignment rather than merely naming a random seed.

Table E.1: Released SIPaKMeD split manifests and hashes.

Protocol	Seed	File	SHA-256
P1	17	sipakmed_p1_seed17.csv	28abd1605d72d888f561cecbc4d987b0dafff671653c37d0460bccac8d3c0cac
P1-prime	17	sipakmed_p1_prime_seed17.csv	611fcb7e6ed98e66a86f9cbdbcb27644850e15d93a4143249dc611d00364d784
P2	17	sipakmed_p2_seed17.csv	488077976eb9397b2a772597e4453e2a31e809d67cb1a21595ece5b143b35262
P3	17	sipakmed_p3_seed17.csv	797a4c0d73fd0722aebec95a14cf798ac3cf94575ac7e93812a85b738df0ea14
P1	29	sipakmed_p1_seed29.csv	adef646d2edabf02bf92a00a7a8e68eefee876159ee2d43d1ab4c25fc80e1e75
P1-prime	29	sipakmed_p1_prime_seed29.csv	72f36aa2713e5dbd4a7d16f471ec466f88dacefbf597d5831bf3b409cc9cb256
P2	29	sipakmed_p2_seed29.csv	4088e6c4b03fca9e16ecb7db57b221e1769e09a5bf6f9c2f36b52519332aed21
P3	29	sipakmed_p3_seed29.csv	3722ec8802c1bbfdf4fe7d9aad2ecd99be4ca66da83add1e331f6d816ed936cc
P1	43	sipakmed_p1_seed43.csv	d5087a1636cceebd8943accfbeb2129046772df01276a693b202f02340846beb
P1-prime	43	sipakmed_p1_prime_seed43.csv	a7417cb3eeccbc4629bd212afffc2c3f42045e9d864f10b232b2c1f15e3a4b1b
P2	43	sipakmed_p2_seed43.csv	bd1a079eadacd51adbeb0dc169d0bcca5cf3b011d3d6b71bb98e4fa71988a62f
P3	43	sipakmed_p3_seed43.csv	877f62465214f160aaa2776bd4ba92c2054c78d744261ae9ada9a4ce66343de8
P1	71	sipakmed_p1_seed71.csv	5148ddbc11d90a927f6cf7d9bc5213d0b1431b9d80cb58fa0d870de76c889198
P1-prime	71	sipakmed_p1_prime_seed71.csv	21ce3cab719badee5dad0bb192c2e55e695fe44d35d58c0a9c6921e4aa7c5b3a
P2	71	sipakmed_p2_seed71.csv	52095a0a39a27f439d64646ed50b07f15bfc729dee195a42f0dcb4c4739d0e25
P3	71	sipakmed_p3_seed71.csv	bbf975f39f790159dcc903bb3d53eb87968e6d612066bfe0cd5c1e8fb6cc860e
P1	101	sipakmed_p1_seed101.csv	1dadec3ecdbce04080b796d735ca7ecfe84cc4b74342d8480326b7c95dd9ff44
P1-prime	101	sipakmed_p1_prime_seed101.csv	d9e096b00642b40eb979f3d75e827f6929e15d0c62a3a9a3738eb4490526185f
P2	101	sipakmed_p2_seed101.csv	178418b059a5c924b305ca5a05472b2076f3423626538e27416ad39f6e1f891f
P3	101	sipakmed_p3_seed101.csv	269ac8e1badbad7c11a5a45cd72b2c2d17ac0fa2c09cb0fa007b45ce51bbef90

F

Git decision chronology

The chronology below is descriptive provenance for major decisions and corrections. The authoritative ordering remains the Git object database. In particular, the first entry is the original preregistration commit.

Table F.1: Complete repository decision chronology through the frozen result set.

Commit	Date	Recorded change
67e8ded	2026-08-22	preregister confirmatory analysis
8ebc6d5	2026-08-22	build leakage verification pipeline
12c9de7	2026-08-22	add parent identity probe and complete source checksums
4ba6a96	2026-08-22	make image provenance recovery independent
1dbd6ac	2026-08-22	separate provenance recovery stages
c551f2d	2026-08-22	add matched classification harness
14a1569	2026-08-22	add near duplicate audit and P3 grouping
79b798a	2026-08-22	amend SIPaKMeD class order before analysis
70b5ffa	2026-08-22	add bounded GPU probe launcher
4aa3e60	2026-08-22	fail closed on image provenance confidence
c18d393	2026-08-22	validate Cx22 against declared source images
f7ca331	2026-08-22	add bounded matched evaluation launcher
318120f	2026-08-22	summarize matched runs in one correction family
8f6f76e	2026-08-22	complete preregistered probe ablations
680fa53	2026-08-22	record reproducible model run metadata
0c5166e	2026-08-22	checkpoint exhaustive provenance matching
ea053e9	2026-08-22	verify structural protocol ordering
65b0230	2026-08-22	export fixed protocol split assets
3261dc0	2026-08-22	render fail-closed verification contract
79e5b31	2026-08-22	require named provenance evidence in report
cded044	2026-08-22	measure literal cross-split pixel overlap
55708d4	2026-08-22	keep training commands free of analysis-only imports
6369698	2026-08-22	generate self-contained thesis figure
da548a4	2026-08-22	bound clean reproduction matcher to three workers
83d56af	2026-08-22	verify published class crop counts
03e7442	2026-08-22	document fail-closed release workflow
be46a53	2026-08-22	separate canonical and audit provenance maps
c617374	2026-08-22	bound provenance image cache to class working set
4c6282e	2026-08-22	use multiscale provenance matching only as fallback
67892b7	2026-08-22	align provenance validation with registered criteria
268e56a	2026-08-22	build explicit section 11 acceptance ledger
438fa3f	2026-08-22	measure duplicate detector precision and recall
62a51ef	2026-08-22	record virgin-host reproduction evidence
dd0a56e	2026-08-22	generate five-seed parent probe figure
b90805a	2026-08-22	reproduce P3 splits and pixel overlap end to end
82e541d	2026-08-22	add matched protocol results figure
adf5539	2026-08-22	document full matched classifier matrix
bf8a4ec	2026-08-22	enforce paper table reporting contract
ab40681	2026-08-22	separate Figure 2 control labels
a72fb5f	2026-08-22	record split and image library versions
4da9f62	2026-08-22	pin CUDA 12.8 training environment
f936802	2026-08-22	verify classification matrix integrity
157b882	2026-08-22	report secondary ordering violations

Commit	Date	Recorded change
10fb2a3	2026-08-22	pin reproducible split runtime
970c74e	2026-08-22	gate acceptance on matrix integrity
8ebb2b9	2026-08-22	simulate exact P1 allocation rule
525df1d	2026-08-22	report architecture ranking stability
1c2da7f	2026-08-22	record preregistered provenance and five-seed results
2c5d5c3	2026-08-22	record post-result acceptance evidence
d347b07	2026-08-24	preregister Cx22 segmentation rescore
a58f72f	2026-08-24	add resumable Cx22 segmentation extension
635e660	2026-08-24	decode Cx22 instance labels with bounded memory
2fdb136	2026-08-24	refresh Cx22 controller heartbeat during epochs
d66dba4	2026-08-24	add memory-mapped Cx22 training cache
a18ddef	2026-08-24	preregister mixed-precision Cx22 execution
f10c9c5	2026-08-24	enable preregistered mixed precision for Cx22
48c9b65	2026-08-28	stabilize Cx22 MAnet training in FP32
373276a	2026-08-28	enforce deterministic CUDA workspace for Cx22
a9594b3	2026-09-06	remove redundant Cx22 instance runtime overhead
01c4c31	2026-09-06	stage Cx22 checkpoints off network storage

G

Exact command cookbook

G.1 Local checks

```
uv sync --extra dev
uv run pytest
uv run ruff check .
uv run cytofair --help
```

G.2 Recover and verify structural evidence

```
uv run cytofair recover-sipakmed data/extracted/sipakmed --workers 3
uv run cytofair recover-cx22 data/extracted/cx22 \
  --image-root data/extracted/ccedd/png
uv run cytofair recover-cric data/raw/cric/classifications.csv
uv run cytofair recover-herlev data/extracted/herlev

uv run cytofair audit-duplicates \
  provenance/sipakmed_parent_map.parquet \
  --hashes results/duplicates/sipakmed_hashes.parquet \
  --pairs results/duplicates/sipakmed_near_duplicate_pairs.csv \
  --output provenance/sipakmed_parent_map_p3.parquet

uv run cytofair make-splits \
  provenance/sipakmed_parent_map_p3.parquet \
  --dataset SIPaKMeD --output splits/generated

uv run cytofair pixel-overlap \
  provenance/sipakmed_branch_b.parquet --dataset SIPaKMeD
```

G.3 Launch the probe and classifier matrices

```
python scripts/launch_probe_matrix.py \
  provenance/sipakmed_parent_map_p3.parquet results/probe \
  --max-gpus 3 \
  --modes rgb grayscale stain_normalized background_only highpass

```

```
python scripts/launch_classification_matrix.py \
  --dataset-map SIPaKMeD=provenance/sipakmed_parent_map_p3.parquet \
  --output results/classification --max-gpus 3 \
  --architectures resnet18 resnet50 densenet121 \
    efficientnet_b0 mobilenet_v2 vit_b_16 \
  --protocols P1 P1-prime P2 P3
```

G.4 Materialize and launch Cx22

```
cytofair materialize-cx22 provenance/cx22_parent_map.parquet \
  --labels /path/to/extracted/cx22 \
  --assets /path/to/cx22_assets \
  --output provenance/cx22_model_map.parquet

cytofair audit-duplicates provenance/cx22_model_map.parquet \
  --hashes results/cx22_duplicates/hashes.parquet \
  --pairs results/cx22_duplicates/pairs.csv \
  --output provenance/cx22_model_map_p3.parquet

cytofair cache-cx22 provenance/cx22_model_map_p3.parquet \
  --cache /path/to/cx22_assets/cache256 \
  --output provenance/cx22_cached_model_map_p3.parquet

nohup python scripts/launch_cx22_segmentation_matrix.py \
  --map provenance/cx22_cached_model_map_p3.parquet \
  --output results/cx22_segmentation --max-gpus 3 \
  > results/cx22_segmentation/controller.log 2>&1 < /dev/null &

python scripts/check_cx22_segmentation.py results/cx22_segmentation
```

H

Artifact index and evidence status

Table H.1: Principal repository artifacts.

Artifact	Status	Purpose
`PREREGISTRATION.md`	Frozen plan	Endpoints, seeds, protocols, exclusions, corrections, amendments.
`data_sources.json`	Versioned	Owner URLs, filenames, sizes, hashes, and terms notes.
`provenance/*.parquet`	Complete derived maps	Observation-to-parent identities, match diagnostics, boxes, hashes, and P3 components.
`splits/*.csv`	Versioned interface	Exact per-seed assignments joined by crop ID.
`results/classification/*.json`	120 accepted	Classifier metrics, environment, split hash, and history.
`results/probe/*.json`	30 accepted	Mechanism probe metrics, chance, environment, and history.
`results/cx22_segmentation/*.json`	175 accepted	Semantic/instance metrics, manifests, runtime, and history.
`results/attempts/`	Excluded, retained	Twelve incompatible split-library runs and three smoke tests.
`reports/generated/acceptance_contract.*`	Current	Machine-readable PASS/FAIL/PENDING gates.
`scripts/reproduce_table1.sh`	Implemented	GPU-free raw-download-to-structural-table workflow.
`scripts/verify_clean_clone.sh`	Implemented, unattested	Non-local clone wrapper and optional virgin-host attestation.
`reports/lab_paper_self_audit.md`	Pending	Nine-record title/protocol crosswalk requiring source-paper checks.
`reports/figure1_reader_test.md`	Pending	Independent reader protocol and blank response record.

I

Complete derived-result ledgers

These tables expose every structural seed, literal-overlap seed/protocol, registered classifier contrast, registered Cx22 contrast, and rank-stability result. The model run ledgers in Appendices A–C are the underlying run-level evidence; this appendix is the complete derived statistical layer.

I.1 Structural and literal-overlap seeds

Table I.1: Complete structural seed ledger.

Dataset	Seed	P1 parent straddle (percent)	P1 test parent in train (percent)	P2 straddle (percent)	P3 straddle (percent)
SIPaKMeD	17	45.549	92.222	0.000	0.000
SIPaKMeD	29	45.445	92.840	0.000	0.000
SIPaKMeD	43	44.203	92.469	0.000	0.000
SIPaKMeD	71	47.101	93.210	0.000	0.000
SIPaKMeD	101	45.859	92.346	0.000	0.000
Cx22	17	34.241	85.985	0.000	0.000
Cx22	29	32.685	80.682	0.000	0.000
Cx22	43	35.214	89.773	0.000	0.000
Cx22	71	34.825	86.742	0.000	0.000
Cx22	101	33.074	82.955	0.000	0.000
CRIC	17	94.750	99.827	0.000	0.000
CRIC	29	95.000	100.000	0.000	0.000
CRIC	43	94.250	100.000	0.000	0.000
CRIC	71	92.500	100.000	0.000	0.000
CRIC	101	95.250	100.000	0.000	0.000
Herlev	17	0.000	0.000	0.000	0.000
Herlev	29	0.000	0.000	0.000	0.000
Herlev	43	0.000	0.000	0.000	0.000
Herlev	71	0.000	0.000	0.000	0.000
Herlev	101	0.000	0.000	0.000	0.000

Table I.2: Complete SIPaKMeD literal source-pixel overlap ledger.

Protocol	Seed	Cross-split pairs	Overlapping pairs	Pair overlap (percent)	Test crops overlapping train	Test-crop overlap (percent)
P1	17	4294	673	15.673	411	50.741
P2	17	0	0	0.000	0	0.000
P3	17	0	0	0.000	0	0.000
P1	29	4153	695	16.735	415	51.235
P2	29	0	0	0.000	0	0.000
P3	29	0	0	0.000	0	0.000
P1	43	4222	749	17.740	443	54.691
P2	43	0	0	0.000	0	0.000
P3	43	0	0	0.000	0	0.000
P1	71	4240	705	16.627	418	51.605
P2	71	0	0	0.000	0	0.000
P3	71	0	0	0.000	0	0.000
P1	101	4199	724	17.242	444	54.815
P2	101	0	0	0.000	0	0.000
P3	101	0	0	0.000	0	0.000

I.2 Classifier contrasts

Table I.3: Complete SIPaKMeD classifier contrast ledger. Secondary tests have no confirmatory BH q-value.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	densenet121	P1	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.985, 4.747]	0.9922
primary	efficientnet_b0	P1	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.668]	0.6215
primary	mobilenet_v2	P1	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.442, 2.365]	0.6256
primary	resnet50	P1	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.057, 4.950]	0.6216
primary	densenet121	P1-prime	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.932, 4.747]	0.9922
primary	efficientnet_b0	P1-prime	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.610]	0.6215
primary	mobilenet_v2	P1-prime	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1-prime	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.426, 2.442]	0.6256
primary	resnet50	P1-prime	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1-prime	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.120, 4.950]	0.6216
secondary	densenet121	P2	P3	5	-0.667	0.691	paired_t	0.0971	–	cohens_dz	-0.965	[-3.266, -0.725]	0.0863
secondary	efficientnet_b0	P2	P3	5	-0.716	1.943	paired_t	0.4562	–	cohens_dz	-0.369	[-2.708, 0.467]	0.6101
secondary	mobilenet_v2	P2	P3	5	-1.037	1.376	paired_t	0.1673	–	cohens_dz	-0.753	[-3.329, 0.075]	0.4329
secondary	resnet18	P2	P3	5	-0.346	1.189	paired_t	0.5510	–	cohens_dz	-0.291	[-1.516, 1.255]	0.7941
secondary	resnet50	P2	P3	5	0.519	0.661	paired_t	0.1545	–	cohens_dz	0.784	[0.361, 2.534]	0.2922
secondary	vit_b_16	P2	P3	5	0.025	2.345	paired_t	0.9823	–	cohens_dz	0.011	[-3.338, 0.817]	0.2169

I.3 Cx22 contrasts

Table I.4: Complete Cx22 contrast ledger. Semantic and instance primary families were corrected separately by task.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	deeplabv3plus_resnet50	P1	P2	5	0.258	0.418	wilcoxon	0.3125	0.8458	rank_biserial	0.600	[-0.200, 1.000]	0.0338
primary	fpn_resnet34	P1	P2	5	0.010	0.468	paired_t	0.9625	0.9625	cohens_dz	0.022	[-1.157, 1.799]	0.9413
primary	manet_resnet34	P1	P2	5	0.065	0.450	paired_t	0.7612	0.8458	cohens_dz	0.146	[-0.653, 3.986]	0.2308
primary	unet_resnet34	P1	P2	5	0.297	0.767	paired_t	0.4353	0.8458	cohens_dz	0.387	[-0.840, 1.537]	0.7640
primary	unetplusplus_resnet34	P1	P2	5	0.188	0.406	paired_t	0.3582	0.8458	cohens_dz	0.464	[-0.447, 7.482]	0.4491
primary	deeplabv3plus_resnet50	P1-prime	P2	5	0.193	0.382	wilcoxon	0.6250	0.8458	rank_biserial	0.333	[-0.600, 1.000]	0.0411
primary	fpn_resnet34	P1-prime	P2	5	0.085	0.549	paired_t	0.7474	0.8458	cohens_dz	0.154	[-1.157, 2.035]	0.5968
primary	manet_resnet34	P1-prime	P2	5	0.102	0.479	paired_t	0.6583	0.8458	cohens_dz	0.213	[-0.694, 4.502]	0.2960
primary	unet_resnet34	P1-prime	P2	5	0.325	0.817	paired_t	0.4240	0.8458	cohens_dz	0.398	[-0.842, 1.506]	0.7316
primary	unetplusplus_resnet34	P1-prime	P2	5	0.206	0.409	paired_t	0.3233	0.8458	cohens_dz	0.503	[-0.375, 6.594]	0.4512
primary	maskrcnn_resnet50_fpn	P1	P2	5	0.509	0.581	paired_t	0.1220	0.2440	cohens_dz	0.875	[0.516, 2.175]	0.2370
primary	maskrcnn_resnet50_fpn_v2	P1	P2	5	0.630	1.204	paired_t	0.3068	0.3116	cohens_dz	0.523	[-0.568, 2.360]	0.7974
primary	maskrcnn_resnet50_fpn	P1-prime	P2	5	0.466	0.340	paired_t	0.0377	0.1508	cohens_dz	1.368	[0.769, 4.031]	0.7993
primary	maskrcnn_resnet50_fpn_v2	P1-prime	P2	5	0.586	1.133	paired_t	0.3116	0.3116	cohens_dz	0.518	[-0.568, 2.651]	0.4050
secondary	deeplabv3plus_resnet50	P2	P3	5	0.546	0.534	paired_t	0.0843	–	cohens_dz	1.022	[0.461, 3.165]	0.6257
secondary	fpn_resnet34	P2	P3	5	0.558	0.346	paired_t	0.0226	–	cohens_dz	1.613	[0.738, 7.765]	0.6155
secondary	manet_resnet34	P2	P3	5	0.339	0.569	paired_t	0.2537	–	cohens_dz	0.596	[-0.284, 2.625]	0.8303
secondary	unet_resnet34	P2	P3	5	0.221	0.579	paired_t	0.4421	–	cohens_dz	0.381	[-0.533, 6.386]	0.1888
secondary	unetplusplus_resnet34	P2	P3	5	0.329	0.299	paired_t	0.0696	–	cohens_dz	1.101	[0.722, 2.785]	0.1372
secondary	maskrcnn_resnet50_fpn	P2	P3	5	0.473	1.435	paired_t	0.5021	–	cohens_dz	0.330	[-1.325, 1.664]	0.2311
secondary	maskrcnn_resnet50_fpn_v2	P2	P3	5	0.761	1.006	paired_t	0.1661	–	cohens_dz	0.756	[-0.093, 3.627]	0.1440

I.4 Rank stability

Table I.5: Complete secondary P1-versus-P2 rank-stability ledger.

Task	Arch.	Seeds	Kendall tau	95 percent CI	p	Reordered pairs	Tie changes
classification	6	5	0.467	[0.071, 0.733]	0.2722	4/15	0
semantic segmentation	5	5	0.800	[0.600, 1.000]	0.0833	1/10	0
instance segmentation	2	5	1.000	[-1.000, 1.000]	1.0000	0/1	0

J

Excluded attempt ledger

No row below was excluded for its score or direction. The twelve twenty-epoch rows belong to the incompatible scikit-learn 1.7.2 split matrix; all were excluded when the entire matrix restarted. The three one-epoch rows are execution smoke tests that were never eligible for confirmatory summaries.

Table J.1: Complete excluded-attempt ledger. Metrics are shown only to demonstrate that exclusion was rule-based, not score-based.

Reason	Architecture/mode	Protocol	Seed	Epochs	Train	Test	Recorded metric (percent)	Eligible
split-library failure	resnet18	P1-prime	101	20	3223	810	98.395	No
split-library failure	resnet18	P1-prime	17	20	3214	810	96.914	No
split-library failure	resnet18	P1-prime	29	20	3237	810	97.901	No
split-library failure	resnet18	P1-prime	43	20	3215	810	97.284	No
split-library failure	resnet18	P1-prime	71	20	3236	810	98.148	No
split-library failure	resnet18	P1	101	20	3239	810	97.654	No
split-library failure	resnet18	P1	17	20	3239	810	97.901	No
split-library failure	resnet18	P1	29	20	3239	810	97.037	No
split-library failure	resnet18	P1	43	20	3239	810	96.914	No
split-library failure	resnet18	P1	71	20	3239	810	96.420	No
split-library failure	resnet18	P2	17	20	3214	835	96.647	No
split-library failure	resnet18	P2	29	20	3237	812	95.936	No
smoke test	resnet18/rgb	–	17	1	848	848	0.590	No
smoke test	resnet18	P1	17	1	3239	810	95.556	No
smoke test	vit_b_16	P1	17	1	3239	810	95.679	No

K

Reading guide for future maintainers

Start with the acceptance contract, not the paper prose. If a gate fails, follow its evidence path into the structured output. Regenerate a summary rather than editing a number. If a library upgrade changes a split, do not overwrite the published CSV: create a new version and preserve the previous hash. If new data supply patient IDs, introduce a stronger protocol rather than relabeling source-disjoint as patient-disjoint. If a model is rerun, retain its per-sample predictions and checkpoint so confusion matrices and qualitative panels become reproducible first-class artifacts.

The simplest invariant is also the most important: an observation may be the tensor consumed by a network, but it is not automatically the independent unit represented by the test set.

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